Field instrumentation

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1 Objective, target group and provision

1.1 Objective

This document states functional and technical requirements related to field instrumentation components and systems part of automation technology competence area.

TR3031 defines the automation technology competence area and states general requirements.

Technical requirements for junction boxes, glands, cables, cable trays, cable ladders and earthing are stated in TR3023 and TR3024.

Technical requirements for Remote I/O cabinets and field termination cabinets are stated in TR3034.

Technical requirements for pressure relief valves, pilot operated relief valves and bursting discs are stated in TR3014.

Technical requirements for HVAC actuators are stated in TR1562.

Technical requirements for vibration systems on rotating machinery are stated in TR3132.

Actuator control panel typicals for on/off valves are given in TR0038.

Life Cycle Information (LCI) such as document formats and document types are stated in TR2381.

GL3032 (Hold) is the field instrument guideline for selection between the various measurements principles. This information is extensive and shall be further discussed with Company and equipment Suppliers before a decision on the most appropriate measurement technology is made.

1.2 Target group

The target group for this TR is:

- System and discipline responsible personnel within HSE, automation, operation and maintenance, process control, IT, security and safety systems, project management, drilling and well technology
- All personnel involved with design, operation, maintenance or modifications of plant technical systems.
- Suppliers and Vendors

1.3 Provision

This document is provided for in “Automation, Technical Requirements and Standards” (TR3030).

2 General requirements

2.1 Standardisation

Field instrumentation components shall be standardised throughout the installation to a limited number of types, and preferably one manufacture for each component type unless otherwise accepted by the Company. Standardization requirements shall not eliminate the possibilities of utilizing new technology where appropriate.

Technical design solutions shall be uniformed and a limited number of variants shall be used.

Field instruments installed outside of hazardous areas should be of the same type as those installed inside hazardous area for the purpose of standardisation, unless their quantity may justifies different stock.

Only one make of instrument tubing compression fitting and sealing compounds shall be used throughout the installation.

2.2 Engineering units

The following engineering units shall be used, unless alternate units are stated by project specific documents:

Table 1 - Engineering units

Physical property	Unit
Pressure	bar, mbar, barg, bara where **Bara = Absolute pressure** is zero referenced against a perfect vacuum, so it is equal to gauge pressure plus atmospheric pressure. **Barg = Gauge pressure** is zero referenced against ambient air pressure, so it is equal to absolute pressure minus atmospheric pressure. **Bar or mbar = Differential pressure** is the difference in pressure between two points.
Level	mm, % for indication
Volume flow	m^3/h (flowing condition), Sm^3/h (standard condition according to ISO 1000)
Mass flow	kg/h
Temperature	°C
Other	According to ISO 1000

2.3 Utilities

2.3.1 Instrument air supply

Instrument air for field instruments including accessories and pneumatic control devices shall be segregated from other consumers (e.g. plant air), and have priority in the air distribution network.

Nominal air distribution pressure shall be 7 barg. Maximum pressure 12 barg and minimum pressure 5,5 barg, measured at consumer location.

The compressed air system shall include receiver and distribution system. The system shall have enough capacity to serve consumers without fluctuations in supply pressure and provide for a minimum of 20% future users.

Each consumer shall be supplied with a separate filter regulator and isolation valve. Air manifolds shall be provided with a drain isolation valve at lowest point.

The dew point of the gas supplied shall be at least 10 °C below the lowest ambient temperature present at the plant. The dew point shall be based on maximum operating pressure.

The supply shall be basically free from contaminations in form of hydrocarbons and solids. Quantity of solid particles should be less than 0.1 g/m^3 and no particle shall be greater in diameter than 3 µm.

For additional guidelines refer to ANSI/ISA 7.0.01, “Quality Standard for Instrument Air.”

2.3.2 Hydraulic supply

The plant shall supply hydraulic fluid for hydraulic control devices. The system shall contain supply and return distribution to each consumer.

The hydraulic system including distribution shall have enough capacity to serve consumers without fluctuations in supply pressure, and shall be equipped with central accumulator capacity to activate all connected consumers minimum once without use of any hydraulic power source. The hydraulic reservoir shall be sized such that, when all actuators are driven, the tank shall not be less than 25% full. When the actuators are returned, the reservoir level shall not be greater than 75%. Provision for expansion of central accumulator capacity shall be included. Measures for avoiding humidity in the system shall be included. The hydraulic system distribution should be designed to allow for flushing of tubing to end devices, not including valve control panel and actuator.

The nominal hydraulic distribution pressure shall be 200 barg. The maximum pressure shall be 210 barg and the minimum pressure 160 barg, measured at consumer location.

Hydraulic supply requirements for special applications, e.g. reservoir downhole valves, shall be derived from case by case studies, per the project requirements.

The supply system for hydraulic control panels for valves shall be independent and segregated from other hydraulic supply systems such as for doors, hatches and winches to avoid cross contamination of the control panels.

The hydraulic fluid shall have cleanliness as ISO4406 code of --/15/12 or better.

2.3.3 Electrical supply

2.3.3.1 General

Requirements stated below apply for facilities according to TR3021 and TR3022, as applicable. For other locations local regulations shall apply. Standard line voltages may differ (e.g. for Canada and GoM, line AC voltage is 120VAC, 60Hz).

Table 2 -Electrical supply for field equipment

Components	Supply
Instrument panels in LER	230 V AC 50 Hz (standard) or 24 VDC
Field instruments	24 V DC (standard) or 230 VAC 50 Hz

Components	Supply
Instrument field panels	24 V DC (standard) or 230 VAC 50 Hz

A minimum of two independent and redundant power supplies shall be used to power instrument systems. Non-critical field instruments with external power may be powered from a single supply 230 VAC 50 Hz.

24 VDC shall be derived from the 230 VAC supply.

Critical systems/safety related systems shall be supplied with dual power (2 X UPS), including dual DC power supplies within user equipment cabinets. TR3124 is refered to for UPS requirements.

DC power supplies should be of hot-swap design.

Segregation, redundancy, and individual isolation shall be ensured throughout the 230 VAC / 24 VDC distributions.

Special attention shall be paid to inrush current of power supplies.

2.3.3.2 24 V DC Distribution System

24 VDC supply shall be floating and adjustable within the range of 24 - 28 VDC.

If earthing of a pole of 24 VDC voltages is required, the Instrument Earthing (IE) system shall be used.

Earth fault and power fault monitoring shall be implemented in the 24 VDC distribution systems.

All power supplies shall be equipped with a common alarm contact (open contact in case of trip).

2.3.3.3 Spare Capacity

All power supplies shall have at least 30% spare capacity with the system fully equipped and at worst working condition.

All 24 V DC distributions shall contain MCBs with necessary numbers of circuits (10% spare to be included).

Input / Output cards and individual field instruments shall generally use fused power feeds.

2.3.3.4 Other electrical sources

Use of other electrical sources, e.g. batteries for wireless instruments, shall be qualified for the application.

3 Design requirements

3.1 General

Instruments that can not be justified to cover a defined need or use shall be avoided.

Local gauges shall be provided facilitate local start-up and monitoring and to assess basic process status in the event of control system failure and possible local venting actions.

Field transmitter output signal loop shall be floating.

Field instruments should be loop powered directly from I/O card. Active or passive input should be defined in the project hardware typicals.

Any arrangement of instruments shall allow for the removal of a sensor/detector head while maintaining the integrity of the other sensors, e.g. in addressable systems like fieldbus systems, wireless systems, smoke detectors.

Galvanic isolation barriers shall be used for IS signals. These barriers should have full smart signal transmission capability.

Electrical/electronic equipment in panels shall be protected against hydraulic leakage (e.g. leakage point such as fittings on hydraulic lines should not be placed so that the electronic equipment can be exposed to the fluid leak).

Instrument impulse lines that can be clogged due to high viscosity fluids, hydrates or other factors, shall have the instrument close-coupled to the process pipeline/equipment. Alternatively, the instrument shall be protected with a remote seal, provided that accuracy and response time are acceptable.
Remote seals shall also be used to protect the instrument when the heat required to keep the process fluid in a free flowing state is greater than the design temperature limitation of the instrument.
Care shall be taken to ensure that the seal fluid will operate within the required temperature range and that it is compatible with the process. The seal fluid shall be nonflammable, have low vapour pressure and low thermal expansion characteristics. Should a diaphragm rupture occur, be compatible with and non-contaminating to the process and the operating environment.
Diaphragm seals may present accuracy problems at pressures less than 2 barg and shall not be used if the required accuracy can not be attained.
2/5 way valve manifold functionality shall be available when remote seal is used.

Each pressure instrument with process connection shall be fitted with instrument block /bleed manifold.
Pressure instruments shall be fitted with a 2-way valve manifold. DP pressure instruments shall be fitted with a 5-way valve manifold.

Locking devices for latching in open position should be considered for for instrument block valves when field instrument is part of a safety function (ESD/PSD).

Full functional independence between control and safety devices shall be assured, including vessel/pipeline connections. Common pressure taps for control and safety devices shall not be used.
Common sources may be considered for gamma ray based control and safety function level measurements, with prior Company approval.

Isolation valves for instrument air supply or for instrument hydraulic supply/return shall be clearly marked to identify the consumer served by each valve.

Field located enclosures and panels shall be equipped with drain nipples. Drain nipple should be placed at the lowest point.

Field located enclosures and panels weighing more than 25 kg shall be equipped with lifting eyes.

For requirement regarding position securing system (interlock), reference is made to TR2315.

3.2 Wireless field devices

Wireless field devices are only qualified for monitoring applications. Use of wireless instrumentation for applications other than monitoring shall require prior Company approval

The following requirements shall apply for wireless field devices:

- Maximum latency (time from originator to gateway) should be of up to 40% of the update rate or up to maximum of 3 seconds
- Battery lifetime should be a minimum of 5 years with the specified update rate
- Alarm for reduced battery life time/health should be transferred to SAS
- Radio transmission range should be a minimum of 50 meters

The wireless networks shall have redundant gateways. Each pair of redundant wireless gateways shall be connected to one logic solver.

Existing Wireless HART gateways should handle a minimum of 25- 30 messages per second.
The recommended number of devices connected to one gateway should be limited to 50 sensors. (Consult also Manufacturer recommendations.)

3.3 Environment

Field devices and commodity items shall be designed for operation in the actual ambient and process environment given in the project design basis.

3.4 Heat tracing

3.4.1 General

External heat tracing of instrument impulse lines, level bridles and wetted parts where required, should be electrical. (ref. TR3021 and TR3022.).

The selection of other types of heat tracing method will be subject to the availability of a suitable heating medium and the particular application.

Heat tracing shall be designed and installed so as not to interfere negatively to measurement results. (E.g. temperature and volume measurements of fluids or gas)

Use of pre-insulated heat trace tubing shall be preferred.

Heat tracing that shall be provided for instruments and instrument piping in services have the following characteristics:

1. The contained or conveyed fluid has a freezing point above the project design low ambient temperature;
2. The relevant properties of the fluid would interfere with the instrument response at or near the project design low ambient temperature;
3. Water vapour or hydrate formation within the fluid could condense, solidify, promote corrosion, or otherwise result in faulty instrument response or premature instrument failure; or,

4. Instrument calibration or operation is temperature sensitive.

3.4.2 Glycol Tracing

Glycol tracing of instruments and lead lines shall use tubing and fittings with project specified dimensions and material. Where stainless steel tubing is in contact with insulation, chloride-free insulation shall be specified to prevent stress corrosion cracking of the tubing.

A separate glycol supply shutoff valve shall be installed in the tracer supply line for each instrument. It is recommended that glycol feed and return tubing be run to each instrument in systems of tube bundles from takeoff points on the headers. Each takeoff point should be valved and tagged with the instrument number served by that takeoff.

When the temperature of the tracing glycol exceeds the design temperature of the instrument or connecting lines, then the process liquid may vaporize in the connecting line, or the corrosion rates may increase greatly. One of the following should be done to prevent this:

1. The tracing tubing should be separated from the instrument and/or connecting lines by insulating blocks or pipe insulation.
2. Glycol should be used at a lower temperature.
3. Smaller size tracer tubing should be used..

Where maximum heating effects are desired, the glycol tracing shall be in direct contact with the impulse lines or instrument (heavy tracing). However, the maximum recommended temperature limits of the Manufacturer shall not be exceeded.

3.4.3 Steam Tracing

Unless otherwise specified, low pressure saturated steam shall be provided for use in instrumentation steam tracing as follows:

Table 3 - Instrumentation steam tracing

	Process Design	Mechanical Design
Temperature (°C)	150	175
Pressure (Barg)	2,5	5,0

Steam tracing of instruments and lead lines shall use tubing and fittings with project specified dimensions and material. Where stainless steel tubing is in contact with insulation, chloride-free insulation shall be specified to prevent stress corrosion cracking of the tubing.

A separate steam supply shutoff valve and condensate trap shall be installed in the tracer supply line for each instrument. It is recommended that steam and condensate tubing be run to each instrument in systems of tube bundles from takeoff points on the headers. Each takeoff point should be valved and tagged with the instrument number served by that takeoff.

Steam traps shall be of the inverted-bucket type.

When the temperature of the tracing steam exceeds the design temperature of the instrument or connecting lines, then the process liquid may vaporize in the connecting line, or the corrosion rates may increase greatly. One of the following should be done to prevent this:

1. The tracing tubing should be separated from the instrument and/or connecting lines by insulating blocks or pipe insulation.

2. Steam should be used at a lower pressure and temperature.
3. Smaller size tracer tubing should be used.

Where maximum heating effects are desired, the steam tracing shall be in direct contact with the impulse lines or instrument (heavy tracing). However, the maximum recommended temperature limits of the Manufacturer shall not be exceeded.

3.4.4 Heat Using Process Flow

Radiant heat from the process flow line may be used for protection of instruments where there is continuous flow and the line temperature ranges between 15°C and 65°C.

The process line shall be bared around the location of the instrument and a weatherproof, insulated housing shall be mounted to totally enclose the bare line and instrument.

3.5 Purging Systems

Purges shall be used when the process fluid would migrate through the seal fluid to the instrument or when neither heat tracing nor diaphragm seals provide satisfactory results.

The purge fluid shall be a liquid when the process fluid is a liquid, and shall be a gas when the process fluid is a gas. The purge fluid shall be clean, free of solids, and compatible with and non-contaminating to the process. The temperature of the purge fluid shall not cause a change of state (i.e. flashing, condensation or solidification) of the process or purge fluid.

The purge fluid pressure shall be sufficient to force a smooth, continuous flow through the process impulse lines and into the main process stream. A purge rotameter (purge regulator) shall be installed and adjusted for a continuous uniform purge flow.

The point of entry for the purge into the instrument lines shall be as close to the measurement connection as practical.

If the instrument is a differential pressure type, then a purge rotameter shall be installed in each process impulse line and adjusted so that the rotameters show equal flow. The transmitter zeroing valve in the instrument manifold shall be open for this adjustment.

Purge systems do not always eliminate the need for heat tracing. Certain viscous streams require heat tracing; not only for the instrument and its connections, but also for the line supplying the purge fluid.

It should also be noted that purge systems may also be used in high ambient temperature conditions. For example, downhole well pressure in thermal heavy oil recovery schemes can often be achieved via the use of gas purge systems (commonly referred to as “bubble tubes”).

3.6 Weather protection

3.6.1 General

If ambient conditions such as deluge, sun, wind, snow and sand can adversely affect the field instrumentation or its functionality, then the instrument shall be sheltered by use of weather protection/protective shades, hoods or enclosures.

3.6.2 Instrument enclosure design

Enclosures with electronic equipment, e.g. instrument transmitters, located such that changes in ambient temperatures can cause internal problems such as condensation or freezing, shall be equipped with internal heating elements or dehumidifiers, in order to prevent deterioration and failure of electronic components.

For electric tracing, an electric heater should be installed inside the enclosure. A self regulating temperature controller or a thermostat shall be used. The heater and controller shall be installed in a manner that does not hinder the removal of the instrument for calibration or maintenance. The heating system shall not promote signal interference with the electronic instruments. .

Where glycol or steam tracing is used, heating shall be achieved by means of a tubing coil supplied and fitted inside the enclosure by the housing Supplier. Such coils shall be installed in a manner that does not hinder the removal of the instrument for calibration or maintenance.

Instrument enclosures for transmitters shall be large enough to accommodate the valve manifold.

Bulkhead unions for process gas service impulse tubing should be placed on the bottom of the enclosure. Bulkhead unions for process impulse tubing for liquid service should be placed on the side of the enclosure. Drain/vent from the instrument valve manifold should be through bulk head unions at the bottom of the enclosure.

When transmitters with indication are mounted in enclosures, then the enclosures shall have window for easy reading. Process tubing connections for calibration and maintenance shall be fitted outside the enclosure.

3.7 Local Control Panel (LCP)

3.7.1 General

See TR3131 for more information about requirements and control philosophy for equipment packages.

The preferred choice should be push-button based LCP with status lights / local indicators.

Signalling to and from a push-button based LCP shall be connected to the instrumented system and not directly to or from field equipment.

Interactive touch screen panels may be used based on prior approval from the Company. Environmental and operational aspects need to be addressed.

The LCP shall contain all necessary status information in order to achieve a safe and predictable operation, and means to transfer the control mode to/from the central control room (local control / remote control switch).

The LCP should be fitted with adequate internal lighting. This lighting shall be suitable for the area classification pertaining to the panel location. Activation of the internal panel lighting may be either manual of automatic, linked to the door operation.

LCPs, except server cabinets, should be designed to avoid the need for cooling fans. Special considerations, such as perforation of cabinet/doors and enforced cooling, shall be taken for server cabinets.

On floating facilities, doors shall self lock in the fully opened position.

Swing frames or logic solvers mounted in the door should be avoided. If used however, special consideration should be made to robustness of cable connections and resistance to vibration and physical contact when opening swing frame/doors.

3.8 Electromagnetic compability

Electrical powered devices shall comply with IEC61000 with respect to both electromagnetic emission and immunity.

3.9 Ex protection

Reference: IEC60079

For ex-requirements to fibre optic cables refer to IEC 60079-28.

All equipment shall comply with the requirements of the specific hazardous area where they are installed.

Field Instruments installed in non-hazardous naturally ventilated areas shall comply with a level of explosion protection according to hazardous area division 2 or zone 2 requirements as a minimum.

Preferred EX protection methods shall to maximum extent possible be:
- Ex i for field instruments
- Ex m for solenoid valves

Ex d may be used for field equipment and shall be provided with an Ex e indirect entry.

Ex p should be avoided where possible and shall be provided with an Ex e indirect entry when used.

Ex n equipment may be used in zone 2 areas and in safe areas outside. A “Statement of compliance” shall be delivered by a Notified Body.

Equipment, which shall remain electrical energised after an APS/ESD situation, shall follow requirements specified in TR1055 or TR2237, as applicable.

3.10 Degrees of protection

Minimum degree of protection provided by local control panels, enclosures, instrument housing (i.e. transmitter housing) and junction boxes shall be as follows, in accordance with IEC60529 (international) or NEMA for facilities in North America.

Table 4 - Enclosure protection

Location	IEC60529	NEMA 250
Outdoor areas, naturally ventilated areas and wash down areas	IP 56	4X
Dry indoor areas	IP 20	12
Other areas	IP 44 (IP 54 for indoor areas with water mist)	13/3S

3.11 Material

3.11.1 General

All equipment and materials shall as a minimum be flame retardant and with preferably no halogen content (e.g. no fluoride, chloride, bromide, iodide)

Equipment enclosures located outdoor, in naturally ventilated areas and wash down areas, shall be made of proven sea water resistant material, or be protected by a coating system according to TR0042.

Precautions against galvanic corrosion shall be made.

3.11.2 In-line instruments and instrument wetted parts

The following specific material requirement applies:

- For in-line field instruments and instrument wetted parts, material shall be selected in accordance with TR2000, but shall as minimum be AISI 316 SS.
- Thermowell in carbon steel pipe work shall be as a minimum AISI 316 SS
- AISI 316 SS shall not be used for thermowells above 60°C.
- For sour service, all wetted parts shall be in accordance with ISO 15156 or NACE MR0175.
- All internal wetted parts shall be made from bar, forging, plate or pipe and shall have equivalent or better corrosion resistance as materials used for body and bonnet. Castings according to TR2000 is also acceptable for internals
- Material for control valves shall be in accordance with TR2212.
- Material for on/off actuators shall be in accordance with TR0038
- Material requirements for valve commodity items shall be as follows:
    - Material quality for mounting assembly e.g. brackets, nuts and bolts diameter 10 mm and less shall be AISI 316 SS.
    - Pneumatic and electric components such as indicators, solenoids and limit switches shall be AISI 316 SS as standard.
    - Air reservoir shall be AISI 316 SS.
    - Air filter regulators shall be AISI 316 SS
    - Rupture discs shall be AISI 316 SS
    - Small relief valves shall be AISI 316 SS
    - Framework for control circuits shall be AISI 316L SS
    - Cabinets for control circuits shall be AISI 316L SS

- Boosters shall be in AISI 316 SS

3.11.3 Off-line instruments and instrument tubing

For sour service offline instruments shall be in accordance with ISO15156 or NACE MR0175.

Off-line instrument elements and body shall be AISI 316 SS.

Material requirements for tubing, fittings and instrument valves shall be stated in pipe class sheets in TR2000, but shall as minimum comply with requirements in table 5. Every project shall have specific pipe/tubing class sheets.

Table 5 - Material requirements for instrument tubing, fittings, instrument wetted parts and instrument valves

Atmosphere/ field environment	Service (Note 1)	Material for instrument tubing (Note 3)	Material for other components i.e. instrument wetted parts/fittings/ instrument valves	Maximum temperature to minimise risk of external crevice corrosion or stress corrosion cracking.	Maximum temperature to minimise risk of internal crevice corrosion (Note 3)
Indoor and dry atmosphere or external non-saliferous atmosphere	Hydrocarbon and hydraulic	SS Type AISI 316 SS	SS Type AISI 316 SS	Max 60 °C (Note 2)	N/A
	Instrument air	SS Type AISI 316 SS	SS Type AISI 316 SS	Max 60 °C (Note 2)	N/A
	Sea water	Titanium grade 2	Titanium grade 2 or grade 5 Hastelloy C-276	N/A N/A	85 °C 30 °C
	Fresh water	SS Type AISI 316 SS	SS Type AISI 316 SS	Max 60 °C (Note 2)	N/A
External and saliferous atmosphere	Hydrocarbon and hydraulic	SS Type 6Mo 25 Cr Duplex	SS Type 6Mo SS Type AISI 316 SS Hastelloy C-276	Max 120 °C for SS Type 6Mo Max 110 °C for 25 Cr Duplex Max 60 °C for SS Type AISI 316 SS N/A	N/A N/A N/A N/A
	Instrument air	SS Type 6Mo 25 Cr Duplex	SS Type 6Mo SS Type AISI 316 SS Hastelloy C-276	Max 120 °C for SS Type 6Mo Max 110 °C for 25 Cr Duplex Max 60 °C for SS Type AISI 316 SS	N/A N/A N/A
	Sea water	Titanium grade 2	Titanium grade 2 or grade 5 Hastelloy C-	N/A N/A	85 °C 30 °C

Atmosphere/ field environment	Service (Note 1)	Material for instrument tubing (Note 3)	Material for other components i.e. instrument wetted parts/fittings/ instrument valves	Maximum temperature to minimise risk of external crevice corrosion or stress corrosion cracking.	Maximum temperature to minimise risk of internal crevice corrosion (Note 3)
			276		
	Fresh water	SS Type 6Mo 25 Cr Duplex	SS Type 6Mo SS Type AISI 316 SS Hastelloy C-276	Max 120 °C for SS Type 6Mo Max 110 °C for 25 Cr Duplex Max 60 °C for SS Type AISI 316 SS	N/A N/A N/A

Notes

1. See TR2023 for sour (H_2S/SSC) service restrictions.
2. A maximum temperature for risk of initiation of crevice corrosion will apply. Maximum 30 °C in chlorinated seawater is recommended.
3. Piping class material to be taken into consideration.
4. When using SS316 fittings with 6Mo tubing, maximum hardness of both materials must be taken into account. It shall be ensured that the compression tube fitting material is hardest and that combining SS316 and 6Mo is in accordance with supplier recommendations.

Material for instrument clamps, where there is risk of vibration, shall be fire retardant polymer. Either AISI 316 SS or fire retardant polymer may be used for instrument clamps where there is noo risk of vibration.

3.11.4 Qualifications

Manufacturers of high alloyed materials such as 25 Cr duplex, 6 Mo, Titanium shall require qualification and acceptance by the Company, and shall be listed in TR2000.

3.11.5 Welding and NDE

All welding shall be continuous. Full penetration welding shall be used for stressed components.

Welding repairs or fabrication welding including weld overlays shall be in accordance with the appropriate valve standard. Where the valve standard does not quote any requirements then the following shall apply:

Qualification of welding procedures for weld overlay shall be in accordance with ASME IX. Acceptance criteria shall be the same as for the base material and the testing shall satisfy the standard and the MDS requirements.

Certification of NDE operators shall be in accordance with EN 473 or equivalent. For GoM certification of NDE operators shall be in accordance with ASNT-TC-1A or equivalent

Specific procedures shall be established for all non-destructive examination.

3.11.6 Other items

Table 6 - Material requirements for instrument housing, enclosures and fire&gas detectors

Atmosphere/ field environment	Instrument housing	Fire&Gas detectors	Protective shades	Local Control Panels	Enclosures
Indoor and dry atmosphere or external non-saliferous atmosphere	GRP with polyester resin or SS Type AISI 316 SS	Fire retardant plastic or SS Type AISI 316 SS	NA	Non-corrosive materials like SS Type AISI 316 SS or GRP	GRP with polyester resin
External and saliferous atmosphere	AISI 316 SS	AISI 316 SS	GRP with polyester resin.	AISI 316 SS	AISI 316 SS or GRP with polyester resin

For instrument housing and fire&gas detectors in external and slaiferous atmosphere, use of other materials than 316SS shall require company approval.

Manual call points shall be of non-corrosive materials. Manual call points for external and saliferous atmosphere shall be glass fibre reinforced plastic or stainless steel 316L.

Hinges and locking arrangements shall be of SS Type AISI 316 SS.

3.12 Signal types

3.12.1 Electrical hardwired field devices

The following signal types shall be the preferred choice for electrical hardwired field devices:

Table 7 - Electrical signal types for field instrumentation

Signal	Type
Analogue input	4 – 20 mA with HART protocol
Analogue output	4 – 20 mA with HART protocol
Digital input	• Potential free contact • Proximity switches with line monitoring (e.g. NAMUR interface)
Digital output	24 VDC normal power, minimum 0,5 mA

Industrial network for signal transfer based on fieldbus protocols may be used. This shall be in accordance with IEC 61158. The preferred choices are Profibus PA and Foundation Fieldbus.

Field transmitters and final control elements using Foundation Fieldbus or Profibus PA shall have documented interoperability, e.g. tick-marked.

3.12.2 Electrical wireless field devices

Wireless network for field devices may be used, provided these have a documented quality and functionality equal or better than the project requirements. Communication protocol shall be WirelessHART or ISA100.11a

3.12.3 Pneumatic field devices

The following signals types shall be used for pneumatic field devices

Table 8 - Pneumatic signal types for field instrumentation

Signal	Type
Analogue input	0,2 – 1,0 barg
Analogue output	0,2 – 1,0 barg

All pneumatic signals shall be powered from the plant instrument air distribution system.

3.13 Provision for future modifications

The following spare capacity that shall be provided at the time of plant start-up.

Table 9 - Spare capacity requirements

Item	Spare capacity
Multi-core cables	20 % increase in number of I/O
Air distribution manifold	20 % installed spare branches, each fitted with a valve and plug

3.14 Instrument tubing and fittings

Use of combined manifolds for piping and instruments valves shall be evaluated. Combined manifolds may be used when instruments are direct mounted on or in the immediate vicinity of the pipe/vessel.

Package Suppliers shall terminate hydraulic and pneumatic tubing at skid edge with bulkhead male connectors or unions.

Instrument ball valves shall only be used for on/off operation.

Where several consumers are located close by, an instrument distribution manifold with needle or ball valves for each consumer shall be used.

There shall be a needle valve or ball valve for shut off of air supply located close (approximately 3 meter) to each consumer.

Air and hydraulic distribution manifolds shall be of compact type with full bore.
The manifold source shall have an isolation valve installed.
Each secondary branch smaller shall be supplied with a block valve located at the take-off point from the manifold. Instrument valves shall be of ball, needle or plug type. There shall be a separate drain valve included in the manifold. Special considerations shall be taken to prevent unintended operation.

Tubing shall be seamless and shall be in metric sizes as standard unless otherwise agreed by the project. (e.g. sizes shall be in inches in the GoM)

Table 10 - Instrument tubing sizes

Application		Size
Signal air, impulse tubing, instrument air supply and hydraulic supply to instruments	Max 387 barg	10 mm x 1,5 mm
	Max 535 barg	10 mm x 2,0 mm
Instrument air supply	Max 12 barg	25 mm x 1,5 mm
Instrument hydraulic supply	Max 250 barg	25 mm x 2,5 mm
	Max 389 barg	25 mm x 3,0 mm

All tubing shall meet or exceed the maximum design pressure for each specific service. Maximum allowable pressure ratings shall be in accordance with the tubing and tubing fitting manufacturer published design data.

Standard compression tubing sizes shall be used wherever possible. Other sizes may be used to satisfy special process requirements and application shall be subject to Company approval.

Tube fitting and instrument small valves for hydraulic service shall have parallel pipe thread connections (e.g. BSPP). Packing ring /dowty ring shall be in 316SS or duplex material.

Fittings for N_2 service in hydraulic applications shall have parallel pipe threads (e.g. BSPP).

Tube fittings for other services than hydraulic and N_2 services shall be NPT threads.

Soft seals for instrument needle valves shall be used wherever relevant.

Pre-insulated heat traced tubing shall be the preferred method for all tube installations where heat tracing is required.

Fire resistant, braided flexible hose connections between the tube and the actuator final fitting may be considered when there is risk of high vibration.

3.15 Measurement uncertainty

Uncertainty limits in table 11 with a confidence level of 95% shall apply.

Table 11 - Uncertainty limits

Instrument	Instrument uncertainty at installation point	Instrument uncertainty
Flow measurement - liquid petroleum	+/- 2,0 % of measured value	+/- 0,5 % of measured value
Flow measurement – Natural Gas	+/- 3,0 % of measured value	+/- 1,0 % of measured value
Flow measurements - liquid water	+/- 2,0 % of measured value	+/- 0,5 % of measured value
Flow measurements - steam	+/- 2,0% of measured value	+/- 1 % of measured value
Flow measurements - air	+/- 3,0 % of measured value	+/- 1 % of measured value
Flow measurements – chemicals	+/- 2,0 % of measured value	+/- 0,5% of measured value
Temperature – inline	+/- 0,5 % of measured value	+/- 0,3 % of measured value
Temperature – clamp-on	+/- 1,0 % of measured value	+/- 0,5% of measured value
Pressure	+/- 0,5% of measured value	+/- 0,2 % of measured value
Pressure DP	+/- 0,5% of measured value	+/- 0,2 % of measured value
Pressure gauge	+/- 2% of full scale	+/- 1% of full scale
Level – separators	+/- 20 mm	+/- 5 mm
Level - scrubbers	+/- 20 mm	+/- 5 mm
Level – degassing units	+/- 20 mm	+/- 5 mm
Analysers Oil-in-water	+/- 4 ppm in the range from 0 – 40 ppm +/- 10 ppm in the range from 40 – 100 ppm +/- 10% in the range from 100 – 1000 ppm	+/-10% of measured value

Analysers – O_2	+/- 5 % of full scale	+/- 0,1% of full scale
Analysers - Cl	+/- 10% of full scale	+/- 5% of full scale
Analysers – Salinity/NaCl	+/- 5 % of full scale	+/- 2 % of full scale
Analysers - pH	+/- 0,5 pH units	+/- 0,2 pH units
Analysers – H_2S	+/- 5% of full scale	+/- 2% of full scale
Analysers - Viscometers	+/- 5% of full scale	+/- 2% of full scale
Analysers – sand detector	+/- 5% of measured value	+/- 5% of measured value

4 Installation design requirements

4.1 Equipment location

General requirements to electrical, instrumentation and telecom installation design are specified in TR3023 and TR3024.

Installation typicals recommendations for field instruments are given in appendix A.

Equipment shall be located/installed:

- in accordance to Vendor requirements
- protected against ingress and mechanical damage (especially for panels)
- protected against vibration (vibration-free remote support)
- protected against weather and water jets
- so that it is easy to operate and maintain
- so that display instruments and flashing lights are legible and visible from main access areas or walkways (e.g. loop powered idicators may be beneficial alternative)
- so that it does not interfere with escape routings, walkways, other equipment, pipes etc.
- to facilitate the removal of other equipment for maintenance, for example coolers, motors etc.
- according to layout drawings (XYZ co-ordinates)

Field equipments shall not be supported on pipe work (including tubing), handrails, access ladders or cable ladders. Instruments which are not classified as in-line shall be either pedestal or bracket mounted. Pedestal shall not be mounted on grating.

Instruments shall not use impulse piping and tubing as support. Instrument supports shall not be mounted on process equipments

Field equipments shall not be mounted on blast walls/explosion relieves.

A lifting area shall be considered for heavy instruments.

Equipment located in areas, which do not allow for required maintenance accessibility shall be installed such that the equipment can be rotated, raised or lowered into areas where maintenance can take place without the need for scaffolding.

Sensing elements such as orifice assemblies, thermocouples, etc., together with their respective process connections and isolating valves shall be accessible from normal working locations without use of scaffolding

If applicable the installations shall be arranged so that they can be heat traced and thermally insulated.

All instruments with indication as main function shall be clearly visible from the equipment they serve. Location of the display element shall be between 1,1-1,8 m above ground level and be fully readable.

Instruments which have been mounted inside enclosures or panels shall be identified by tag number plates on the outside and inside.

Sensing elements such as orifice assemblies, thermocouples, etc., together with their respective process connections and isolating valves shall be installed within 0,5m horizontally, and 2m vertically from floor, platforms or walkways. In order to achieve this, platforms shall be extended, or extra platforms shall be provided whenever necessary.

4.2 Field architecture and segregation

The field architecture shall reflect project split-location fabrication and ease final assembly by minimizing hook-up items.

Field instrument and valve signals shall be hardwired directly to Remote Input/Output (RIO) units located locally in different areas. Junction boxes should be avoided, but can be justified if a field device has several signals that connect the same automation system, e.g. PCS, PSD.

Field transmitter signals may be hard wired directly to safe area located Field Termination Cabinet (FTC) if suitable.

4.3 Installation of wireless instrumentation

When installing new wireless instrumentation network, other networks (WLAN) shall be identified and coexistence of the networks shall be verified. Where conflicts exists between networks an evaluation shall be done to have the WLAN operate in the 5 GHz band instead of the 2.4 GHz band, i.e. using the 5 GHz portion of IEEE 802.11n, optionally IEEE 802.11a instead of IEEE 802.11b/g

Factors that may affect the transmission range such as machinery, pipes, walls, vessels and electromagnetic noise shall be identified. The maximum distance between network transmission devices should be 25 meters (82 ft). Consult Manufacturer for additional compatibility and performance factors.

4.4 Tubing installation

4.4.1 General

Instrument tubing shall be installed according to the recommendations of the Manufacturer.

Instrument tubing shall not support connected components.

Instrument tubing sizes less than 16 mm outside diameter shall be supported to field trays or cable ladders.

Trays are not required for internal tubing on components if tubing is sufficiently protected.

Tubing with size equal to or below 25 mm outer diameter, to be fastened to tubing clamps with span max every 60 x tubing diameter, multitube approximately every 2,5 m for vertical runs and 1,5 m for horizontal runs.

Tubing sizes above 25 mm outside diameter shall as a minimum have support every 1,5 m.

Galvanic corrosion between tubing and tubing support system shall be avoided. The tubing clamp shall, when installed, not allow for water/sea water to be accumulated between tubing and tubing clamp on wall. Tubing clamps should be of a self draining type.

Parallel runs of tubing on the same support shall be arranged such that it is possible to have access to every connection point. Fittings should be maximum 0,25 m from support to instrument tubing on hydraulic lines.

Installation into or through panels shall be by use of bulkhead unions or multi cable transits (MCT).

Instrument tubing and cables may be installed on the same field tray for shorter distances (appr. 5 m). Instrument tubing and cables may be routed through the same cable/tubing penetration provided the transit is approved for such use.

All tubing and/or tube fittings which are not connected shall be sealed by use of end-plug / cap of same material as the tubing, fittings.

Test connections, vent, drain and manifold valves shall be available external to applied thermal insulation.

Capillaries of filled systems shall be continuously supported by field trays. Capillaries shall be coiled inside a cable tray or have similar mechanical protection.

Sealing compounds for process services and instrument air services shall require approval from the Company.

Impulse tubing shall be as short as possible and be installed so that gas/liquid pockets are avoided.

Impulse tubing shall have minimum 1:12 slope.

The slope shall be as follows:

Liquid	-	down from the tapping point
Steam and Condensate	-	down from the tapping point
Gases	-	up from the tapping point
Cryogenic	-	up from the tapping point

Contractor shall ensure that vents are placed at the highest point of the installation and drains at the lowest.

Special considerations shall be made where movement of connected equipment can occur.

Multi tube bundles shall have sufficient straight length before end connection.

Field instrument process connections shall be 1/2" NPT minimum.
Field instrument pneumatic connection shall be 1/4" NPT.
Field instrument hydraulic connections shall be 1/2" parallel pipe threaded end fittings (typically BSPP) without adapters to ISO 228-1, sealing surface to DIN 3852, form A, utilising a bonded seal packing in Duplex SS/PTFE.

The orientation of the process tapping point shall be in conjuction with the slope. Slope down to process means tapping up, slope up to process means tapping down but at +/-45 º with reference to six o'clock
For interface to process hook-up see also TR2325 .

4.4.2 Cleaning

All tubing shall be cut by proper tubing cutting tools and de-burred. Tubing shall be blown through with clean, dry air before final installation.

All tubing in hydraulic systems shall be hot oil flushed and cleaned in accordance to TR2323 ch.8, cleanliness level, ISO 4406 17/15/12.

4.4.3 Tubing and distribution manifold labelling

Distribution manifolds shall be tagged.

Tubing for pneumatic supply and hydraulic supply and return shall be marked with the consumer tag number, either at the distribution manifold or at the point of connection to the main line.

Tubing passing through bulkheads/MCT shall be marked with consumer tag number on both sides of the penetration point.

4.5 Thermal Insulation

Capillaries shall be thermally insulated if ambient temperatures can cause significant change in capillary volumes and mismeasurement. Capillaries shall not be heat traced.

Level transmitters in hydrocarbon service shall follow the insulation requirement given in the vessel trim number. However as a minimum the level transmitters shall be thermally insulated.

Level gauges in hydrocarbon service shall be thermally insulated and heat traced without exceptions. Insulation shall include drain and vent valves on the level gauge.
If the vessel is fire insulated, fire insulation of the level gauges itself is not required, however the nozzles to the vessel shall be fire insulated and heat traced.

4.6 Winterisation

Winterisation shall be accomplished by first considering indoor installation, then outdoor installation with suitable instruments and heated enclosure.

Where instrument process tubing is located outside and is prone to freezing or process fluids which may become viscous, prefabricated, traced and insulated tube bundle shall be used.

Instrument process tubing for differential pressure instruments shall have common heating and insulation to assure equal temperature in both lines. Heat traced tube bundles are preferred.

4.7 Temperature instruments

When thermowell is used the temperature transmitter should be mounted directly on top of the thermowell.

Where the access to the measurement point is difficult, the possibility for installing the temperature transmitter inside a RIO-cabinet shall be evaluated. Cable length and influence from electromagnetic noise should be evaluated.

Thermowells shall be mounted in such a manner that the element can be installed and removed from the thermowell for maintenance, without disconnecting the cable.

Thermowells shall be located downstream of flow meter so as not to disturb the flow pattern in the line upstream of the flow meter.

4.8 Level instruments

4.8.1 General

Level transmitters shall be accessible, removable, and capable of isolation in place from the tank exterior, for calibration and maintenance purposes.

Instrument nozzles shall be located at the top and/or side of the vessel, such that negative impacts on the measurement are minimized.

Cage side/side mounted interface for liquid/liquid application shall be designed with three process connections and be jig set from the factory to avoid any installation alignment issues.

All instruments and gauges shall have dedicated nozzles.

Relief/drainage tubing or pipe shall be routed to a safe location according to area requirements.

Definitions for 0% and 100% level shall be equal for all level instruments and glasses.
0% shall be the lowest measurable level and 100% shall be the highest measurable level.

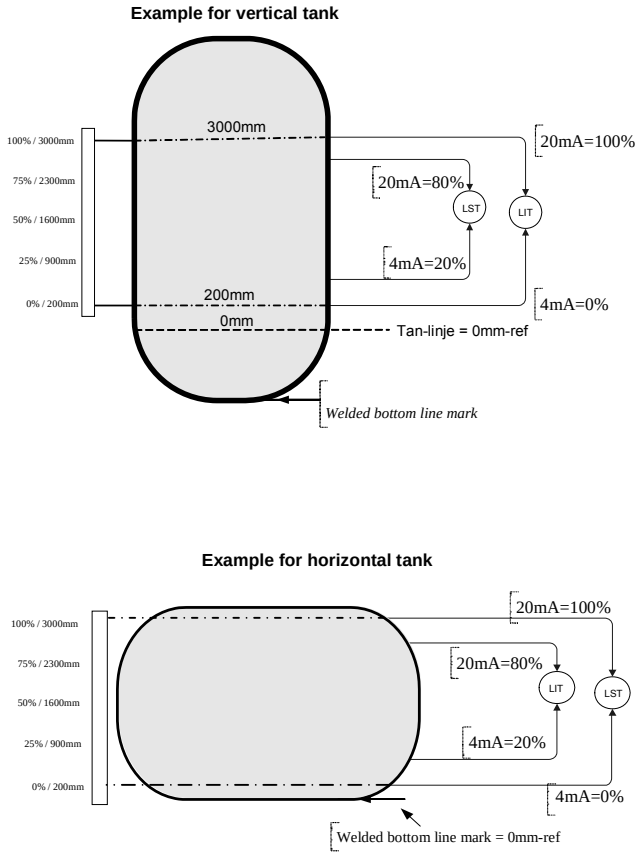
Definitions for 0 mm in horizontal tanks shall be the inside bottom. For vertical tanks, the tangent line shall be 0mm. For side mounted nozzles, 0% and 100% level is measured from the center of the respective nozzles.

Reference point for 0% shall be marked on the vessel.

Level glasses shall as a minimum cover the total range of transmitters for both safety and control.

Vessels containing liquids with different specific gravities shall be fitted with two overlapping level glasses, not a single large one. This is to prevent a liquid seal forming and to allow the various layers to be seen. Visible sections shall overlap by at least 50 mm.

Figure 1: Example of 0 – 100% level



4.9 Flow instruments

Flow meters shall be installed according to recognized standards or according to the recommendations of the Manufacturer. Standards referred to in chapter 5.3 contain additional installation requirements, which should be followed.

Flow meters shall be placed upstream of control valves and other pressure reduction items
Irrespective of design, the flow meter shall not cause flashing or cavitation at any position in the line. Pressure drop shall generally be kept to a minimum and shall not cause an unacceptable restriction to flow unless it is a dedicated flow restriction.

The design shall take into consideration the need for space above/below the flow element for isolation valves, tubing and transmitters.

Ultrasonic meters should not be installed with the transducers positioned at the top or bottom of the meter, in order to avoid adverse effects from gas or solid deposits.
Installation of ultrasonic meters in wet gas (e.g. from 1st stage separators) processes must be carefully evaluated. Both placements of transducers as well as mounting of the meter shall be considered.

Noise from valves or process equipment up or down stream the ultrasonic meter shall be evaluated.

When installing orifice plates correct postion of the drain hole shall be evaluated.

The following upstream and downstream straight pipe lengths should be used for design of flow meter installations:

Table 12 – Upstream/downstream pipe length

Measurement principle	*Upstream pipe length*	*Downstream pipe length*
Orifice	Minimum straight lengths (ISO 5167: part 2, Table 3 column B or Manufacturer recommendation) shall satisfy the measurement accuracy requirements for the application.	
Venturi tubes	Minimum straight lengths (ISO 5167: part 4, Table 1 column B or Manufacturer recommendation) shall satisfy the measurement accuracy requirements for the application.	
V-cone	***3D***	***1D***
Annubar	***20D***	***5D***
Ultrasonic	***20D***	***5D***
Vortex	***20D***	***5D***
Variable Area Meter	***No requirements***	***No requirements***
Electromagnetic Flow Meters	***5D***	***2D***
Turbine Meters	***20D***	***5D***
Coriolis Meters	***No requirements***	***No requirements***

The use of flow straightners or flow conditioners shall be considered, where long upstream pipe lengths such as 20D or greater are required.

The use of a strainer shall be considered upstream of turbine meters to avoid damage to the turbine blades or rotor.

4.10 Fire & Gas equipment

Fire and Gas equipment shall be installed according to Supplier requirement.

Sensor range, cone of vision, sensor direction and angle shall be considered.

The detection system shall be designed and installed in such a way that it can be functionally tested without interrupting the normal plant activities.

For gas detection in ventilation inlets the detector shall be located as close as possible to the inlet to ensure fast detection.

Arrangement shall be made for testing of F&G equipment in order to avoid use of temporary arangement (scaffolding etc).

Flame detectors shall be installed in a way that facilitates cleaning of the optics.

Sensors exposed to weather, should be oriented in a downward facing position, and if possible, not be pointing directly into the sun. Each detector should be protected by drip or splash guards.

All detectors shall be provided with a “pig tail” in such way that the equipment can be moved 2 meters in any direction without disconnecting the cable.

Early warning detectors shall be mounted minimum 0.5 metre away from light sources.

Gas line of sight detectors and flame detectors should be mounted directly on the structure to avoid vibration. If a separate stand is used, then it must be stiffened to avoid vibrations.

The minimum distance from line of sight receiver to any light sources shall be as per Supplier recommendations.

Based on a typical flame detector characteristic, the distance between flame detectors and targets monitored should not exceed 26 m or the Supplier recommended distance.

For flame detectors sensitivity to external impact such as arc welding, burner boom and flare etc. shall be considered in order to avoid activation of sensors in a non fire situation. Special means shall be applied to avoid false triggering from steady state radiation sources

Heat detectors shall have a maximum coverage in naturally ventilated area of approximate 24 m2, maximum distance between sensors 7 m and maximum distance from wall 4,5 m and minimum 0,5 meters away from outside wall or dividing partition.
Heat detectors shall have a maximum coverage in mechanically ventilated area approximate 37 m2. the maximum distance between sensors shall be 9 m and the maximum distance from the wall 4,5 m.

Smoke detectors shall be mounted 100 – 300 mm below the roof, straight below steel beams.
If the detector is mounted on a ceiling or a completely flat roof, then the Supplier’s guideline shall be followed.

Smoke detectors shall have maximum distance between sensors of 11 meters, maximum distance from sensor to bulkhead of 5,5 meters and shall be a minimum of 0,5 meters away from outside wall or dividing partition.

Smoke detectors mounted in areas such as suspended ceilings and raised floors shall have their location indicated by visible tag-plate.

Catalytic and Electrical chemical gas detectors shall be mounted with the head pointing downward.

For gas line of sight detectors, there shall be enough space around the receiver and transmitter to install the alignment equipment.

Gas point detectors not accessible from normal access level shall be equipped with a test tube down to 1500mm above access level for easy access and regular testing of detectors.

A working method for optimization of installation of duct mounted gas line detectors shall be established during detailed engineering. This is in order to avoid condensation on mirrors. Duct penetrations shall be according to the Supplier’s recommendations.

Point gas detectors in air inlets shall be protected against water ingress. Maintenance access for such detectors shall be addressed.

Manual call points shall be mounted 1.5 meters above ground level.

5 Field instruments

5.1 General

This section states requirements for the most frequently used measuring principles. Other types may be used on special applications. For field instruments not described within this document, the design shall be based on recognised international standards or Supplier recommendations.

Instrument performance/accuracy shall be sufficient to fulfil process/unit performance requirements and shall be selected to minimize calibration frequency and maintenance.

The Supplier shall provide data for long-term stability and confidence level for the field device performance.

Field instruments for safety functions shall be documented to be suitable for their intended purpose, in compliance with IEC 61508/61511 or ANSI/ISA S84.01 and the corporate requirements in TR1055 and TR2237, as applicable.

Instrumentation used for hydrocarbon services shall be constructed in such a way that any fault in the primary process barrier will not leak into the main compartment or junction box. If a leakage occurs, it shall not be possible to build up any pressure inside the instruments.

Mechanical switches shall be avoided in favor of analog instruments.

Field instruments and controllers of pneumatic type should generally be avoided.

Transmitters shall have integrated local indicators. Separate local indicators shall be used if physical placement of transmitter complicates accessibility and readability or for operational purposes, (e.g. local control panels or where a local control point is located away from the point of measurement).

Field instrument cable entry shall have ISO threads - size depending on cable size.

Flange connection for inline instruments shall follow piping class, see TR2000.

All in-line flow elements (when part of the process line) shall be flanged for removal from the process line.

Field transmitters and final control elements shall have possibilities for online self-reporting.

All instruments given a tag number shall be delivered with a 316SS tag plate, minimum size 50 x 20mm, with etched or embossed characters of minimum 5 mm high. This tag plate shall contain the tag number only. The plate shall be wired to the instrument with a stainless steel wire.

Nameplate of the field instrument shall as a minimum contain:

- Instrument serial no.
- Instrument model no.
- Protection category and Eex/ATEX certificate no. (if applicable).

5.2 Temperature measurement

5.2.1 General

Technical requirements stated in API 670 shall apply for temperature sensors used in machinery protection systems. Temperature measurements in rotating machinery such as metal temperature in bearings, windings or drain temperature shall be recorded and presented on the SAS system. See TR3132 for further information about instrumentation for machinery systems.

Non-intrusive temperature measurements not requiring a conventional or weld-on type thermowell should be used if the accuracy and response time requirements are met. Non-intrusive elements should always be considered for high vibration or high flow velocity applications.

Non-intrusive temperature transmitters shall be constructed to meet the anticipated vibration levels. They should be easy to mount and have sufficient support strength to mount instruments with dual PT100 RTD elements, wired via a connecting a tube to a common transmitter housing mounted on the clamp.

Non-intrusive temperature transmitters should only be used for line sizes 3 in or larger.

Thermowells should be used for temperature measurements that are not of the clamp-on type.

In addition, thermowells shall be used if any of the following apply:

- operation temperature exceeds 160 degrees centigrade
- pipe wall thickness exceeds 25 mm (for duplex or materials with similar heat conductivity)
- the pipe is manufactured in a material with significantly lower heat conductivity than stainless steel
- sufficient thermal isolation cannot be achieved
- the pipeline is heat traced
- there is not a straight pipe of sufficient length where the temperature is to be measured
- significant heat radiation from the flare or other sources will disturb the measurements

Thermowells shall be used for fiscal measurements, heating medium systems, on heat traced pipelines, HVAC liquid systems, process vessels and safety systems.

Temperature elements not accessible during operation shall for selected critical equipment be installed with dual elements (backup element) and terminated in the field for easy access.

Temperature measurements should be performed by PT 100 RTDs in accordance with IEC 60751.

For temperature measurements above 600 °C thermocouple material Chromel Alumel (type K), in accordance with IEC 60584-1, IEC 60584-2 and IEC 60584-2-am1, should be used.

Temperature sensors shall be enclosed in a 6mm diameter AISI 316 SS sheath, preferably spring loaded to ensure contact with the thermowell.

Where a thermowell has been specified, it should be delivered together with its associated temperature sensor.

Temperature transmitters that accept dual elements and have the possibility to swap elements upon failure (hot back-up) shall be considered.

5.2.2 Thermowells

Thermowell flange arrangement shall be in accordance with TR2325 .

Thermowells shall be of the flanged type, size 1,5 in, normally. For tanks and vessels, and for pressure class 2 600 lb and above, the size shall be 2 in.
For non-critical utility service, thermowells of threaded type (NPT) can be accepted.

The thermowells inner diameter shall be suitable for temperature elements with stem diameters of 6 mm OD.

Thermowell lengths shall be standardised. See Piping Detail Standard TR2325 for technical solution of thermowell.
Also API 551 should be consulted to determine thermowell insertion depth.

Welded thermowell shall be constructed in accordance with Norsok M-601 and all welds shall be liquid penetrant tested.

Thermowell strength calculations (e.g. wake frequency calculations) shall be performed for process hydrocarbon systems according to ANSI/ASME PTC 19.3 TW-2010.

Thermowells for insulated vessels should have extension necks.

5.2.3 Temperature gauges

Bi-metallic or gas filled temperature gauges with 100 mm nominal head diameter shall be used for local indication with 360 degree orientation. Stem diameters shall be 6 mm OD.

Normal temperature shall be at approximately 60% of full range

Temperature gauges with capillary tubing should not be used.

Manufacturer's standard ranges shall be used and dials shall have large, black and easy to read, digits on white background.

5.3 Flow measurement

5.3.1 General

See TR0814 for flow measurements for fiscal applications.

Robustness with regard to variations in operating conditions and fluid properties need to be given due consideration, when selecting a flow meter. Selection of the measurement principle and design of the piping arrangement are critical for performance and measurement quality.

Flow meters shall not be used in applications with higher gas or water content than the Manufacturer accepts for a guaranteed meter performance.

Flow elements shall not be used where possible loss of mechanical components upon failure can endanger downstream equipment (e.g. insertion turbine meters in a compressor inlet).

Evaluation criteria shall as minimum include the following:

- process conditions (pressure, temperature, phase, etc.)
- fluid properties (clean/dirty, dielectric properties, etc.)
- turn down requirements
- uncertainty requirements
- traceability requirements
- available pressure drop

- installation requirements
- acoustique noise / ultrasonique noise from other valves etc.
- relevant piping specification requirements for the flow meter
- operational experience
- maintenance requirements
- verification of meter performance in operation

.
For test and / or inlet separator metering, the design shall be based on an evaluation of the following:

- whether or not the separator is to be used as reference for calibration of multiphase meters.
- the presence of liquid carry-over in gas stream
- avoiding flashing conditions in a liquid flow meter
- bypass arrangements on liquid meters (turbine meters)

Flow elements shall be marked with flow direction, pressure rating and appropriate hazardous area classification.

The flange and piping arrangement shall be in accordance with TR2325 .

5.3.2 Preferred measurement principles

5.3.2.1 Petroleum liquid process flow measurements

Metering principles which shall be considered for petroleum liquid flow measurements are:

- turbine
- ultrasonic
- coriolis
- orifice plate
- v-cone

Other measurement principles may be considered in special cases such as for very high turndown requirements, liquids with high viscosity or measurements on very high or very low flow rates.

5.3.2.2 Gas process flow measurements

Metering principles which shall be considered for gas flow measurements are:

- ultrasonic
- orifice plate
- v-cone
- venturi

In special cases (typically low flow rates) coriolis meters may also be considered.

For applications such as flue gas, where $C0_2$-concentration are high in combination with low pressure and high turndown requirements, finding a good metering principle may be challenging and the metering principle shall be advised by the Company.

5.3.2.3 Water liquid process flow measurements

Metering principles which shall be considered for water flow measurements are:

- turbine
- ultrasonic (clamp-on may be considered)
- coriolis
- electromagnetic
- orifice
- v-cone

Other principles may be considered for special applications such as low flow or very dirty liquids.

5.3.2.4 Chemical liquid flow measurements

Metering principles which shall be considered for chemical injection liquid flow measurements are:

- coriolis
- positive displacement
- turbine meters

Other measurement principles may be considered in special cases.

5.3.2.5 HVAC/Air Flow measurements

A DP flow measurement principle should be used.

The metering principle shall be chosen in conjunction with the HVAC system Supplier.

5.3.2.6 Steam Flow measurements

Metering principles which shall be considered for steam flow measurements are:

- Vortex
- Orifice plate
- Ultrasonic
- Variable area
- Annubar
- Coriolis

5.3.3 Orifice plate

The orifice-plate shall be calculated, manufactured and installed according to ISO 5167 or meet the recommendations of the manufacturer.

The minimum pipe size for orifice plate metering should be 2". If process piping is smaller than 2", then the pipe size shall be increased to 2" for the meter run. Should this not be possible then an integral orifice meter or a precalibrated meter run shall be used.

Orifice plates shall be provided with a handle of sufficient length to extend beyond the pipe flanges. This is to enable the information stamped on the upstream side of the handle to be read.
Information to be stamped on the handle is to include the orifice bore (d), the nominal pipe size (D) the material used and the instrument tag number.

Dual chamber orifice fittings may be used to cover services with wide flow rate ranges.

A third-party verification shall be performed of design and calculation model of multihole restriction orifices used for blowdown applications. Hole tolerances should be verified.

5.3.4 Venturi tubes

Venturi tubes shall be calculated, manufactured and installed according to ISO 5167 or meet the recommendations of the Manufacturer.

5.3.5 V-cone flow meters

The v-cone shall be calculated, manufactured and installed according to Manufacturer recommendations.

5.3.6 Annubar Flow Meter

This type of meter shall only be used on clean gas, steam, clean liquids and air applications.

Annubar flow meters shall not be used for low flow applications where the Reynolds number is less than 10000. The Reynolds number should be above 30000.

Annubar meters should be designed to allow their extraction during operation.

5.3.7 Ultrasonic flow meter

Selection and installation design of ultrasonic meters should be according to AGA 9 for gas applications and to API MPMS, Chap. 5.8 for liquid applications.

The measuring principle shall be the “transit time differential method”.

Clamp-on ultrasonic meters shall only be used for non-critical measurements.

Before a clamp-on meter is chosen, an evaluation of position, process fluid, flow conditions and piping material including isolation and lining material shall be conducted.

5.3.8 Vortex flow meters

Meter can be used on both liquid and gas applications

The meter shall not be used for low flow where Reynolds number ReD <10000 and on low pressure application. Reynolds number should be above 30000.

5.3.9 Variable area flow meters

Variable area flow meters shall only be considered for simple duties, such as purge or seal fluid monitoring etc.

Variable area meters should have metal tubes with flange connections and actual flow indicating scales.

Glass tubes shall only be used on fluids such as clean water, air or nitrogen etc. which in the event of a breakage will not be hazardous to personnel or plant. Screw connection can be applied for these applications.

5.3.10 Electromagnetic flow meters

Electomagnetic flow meters can be used for water applications with a continuous conductivity according to Manufacturer’s recommendations.

5.3.11 Turbine flow meters

The selection and installation design for turbine meters should be according to API MPMS Ch.5.1 and 5.3 for liquid applications and to ISO 9951 for gas applications.

For facilities in North America, AGA 7 should be used for natural gas applications.

Signal amplifiers shall be mounted on or adjacent to the meter.

5.3.12 Coriolis flow meters

Selection and installation design should be according to AGA 11 or API/MPMS Ch.5.1 and 5.6 section 6 and 7 or ISO 10790.

5.4 Pressure measurement

5.4.1 General

Minimum one pressure transmitters installed on piping or equipment protected by a pressure relief device, shall have a measuring range covering the set pressure of the relief device

Pulsation dampers shall be used if pulsations are likely to occur.
Needle valves shall not be used as pulsation dampers.

All pressure instruments shall withstand an overpressure of minimum 130 % of upper range value without the need for recalibration.

Differential pressure instruments shall be able to withstand full static (line) pressure on either of the inputs, with the other input at zero (atmospheric pressure) without damage to the sensor and without the need for recalibration.

Differential pressure instruments equipped with capillaries and remote seals shall have 3 inch flanges and flushing ring equipped with two 1/2" NPT FEMALE plugged connections. Smaller diaphragms shall be considered for high pressure measurements. When used capillary leads shall be kept as short as possible

Diaphragm seals may present accuracy problems at pressures less than 2 barg (30 psig) and shall be used only if the required accuracy is attainable.

Diaphragm seals shall be fitted to pressure instruments in corrosive, viscous or services containing solids, provided that the desired accuracy can be achieved.

Ranges below 50 mbar shall be avoided.

Use of permanent pressure gauges shall be minimized.

Pressure switches shall be avoided.

5.4.2 Pressure gauges

Pressure gauges shall be of the heavy duty liquid filled type with safety glass and rear safety blow out disc.

Gauges with ranges from 0,6 barg, shall have bourdon type element and shall have liquid filled house/case. The measuring element shall generally be a Bourdon tube type unless process considerations dictate the use of a diaphragm seal type to prevent corrosion, plugging etc.

Normal pressure shall be at approximately 60% of full range.

The nominal housing/case diameter for process applications shall as a minimum be 100 mm.

Dual element gauges or differential pressure gauges for measurement of differential pressure should not be used.

5.5 Level measurement

5.5.1 General

Direct vessel mounted instruments with non-moving parts should be used.

The level application shall have compensating measures if variations in process conditions (e.g. temperature, pressure, density) will affect the measurement accuracy.

External bridles shall typically be of 2" to 4" nominal diameter, with 2" side/side flanged process connections and shall be equipped with ¾" flanged vent and drain connections.

An arrangement for in-line calibration and flushing of the instrument shall be included.

Level sensing instruments that can be clogged due to high viscosity, hydrates, foam, scaling, build up, or affected by other factors shall be equipped with flushing possibilities. Flushing possibilities may be manual or automatic(e.g. flushing port for DP).

Measuring principles shall be selected according to application. Evaluation criteria should as minimum include the following:

- non moving parts
- density
- pressure
- accuracy
- temperature
- vessel geometry
- nozzle locations
- clogging
- scale build-up
- foaming, emulsion or turbulence problems
- vibration / level fluttering
- fluid level interface phases
- process inlet and outlet
- installation induced movements (floater, ship etc.)
- top mounted, side mounted or mounted in a cage.
- Maintenance and calibration

The flange arrangement shall be in accordance with TR2325 .

5.5.2 Preferred measurement principles

Gamma and differential pressure shall be the standard level measurement principles for process control and safety applications for two-phase and three phase applications.

Guided wave radar measurement may be considered for two-phase and three-phase applications.

Vibration level switches may be used for high level safety functions. Switch should not be used for low level.

Level instruments such as ultrasonic, capacitance, displacer or conductive type may be used for special applications e.g. homogeneous applications such as lube oil systems.

5.5.3 Local level indicators (gauges)

Level indicators shall cover maximum and minimum operational levels including high/low trip points.

The scale on the level glass or magnetic level gauges shall indicate both [mm] and [%].

Calibration of level instrument and level glass shall be identical, for the same level measurement.

Selection of level gauge type readability shall be evaluated based on the actual fluid process medium. Integral back light for transparent level gauges shall be considered.

Gauges with magnetic indicators should only be used in clean liquid applications, i.e. there should be no risk of the magnetic displacer unit getting stuck or contaminated due to the process fluid.

Magnetic type level gauges may be used for ranges above 1200 mm, eliminating the need for multiple level gauges.

Shrouded covers shall be used to protect glass edges from sandblasting.

Gauge glass type shall allow for cleaning during operation.

Reflex type gauge glass shall only be used on clear mediums. For interfaces (oil/water), the transparent type gauge shall be used. Transparent type gauges shall be equipped with back lights.

Reflex and transparent gauges shall be equipped with glass breakage automatic shut off valves.

Level gauge glasses shall have flanged process connections, and be fitted with gauge valves for maintenance purposes. The gauge valves shall be located so that complete drainage of the gauge is possible.

Simpler solutions may be used on small and non critical vessels.

5.5.4 Differential pressure

Pressure instruments shall be installed with permanent connections for calibration. DP level instruments shall be mounted on dedicated nozzles.

Differential pressure measurement instruments for pressurized vessels shall be of the capillary type, with remote seals. Electronic seals without the use of capillaries may be used on vessels with a low static pressure

Remote seal process connections shall have 3 inch flanges.

Remote seals shall be equipped with a means of flushing, venting and draining of the diaphragm process side. A minimum of two off ½" female NPT connections shall be provided for this purpose.

5.5.5 Capacitance

If there is risk of contamination from wax or scale, then the capacitance meter shall have an automatic cleaning system.

Probes shall not be used for applications where there is a risk of variations in electrical conductivity of the process media.

5.5.6 Guided wave radar

All units shall be capable of level determination for products with a dielectric constant as low as εr =1,4. As well for interface applications, the units shall be able to handle upper products with a dielectric constant as low as 1,4 when the dielectric constant of the lower product is at least higher than that of the upper product by 10 (i.e. upper dielectric constant = 1,4/ lower dielectric constant = 14.

In applications suffering from chalk like sediment, build up, scaling, hydrates, crystallization or condensation the probe shall be suitable for its intended use. Single, rigid or single wire or enlarged co-axial with flushing connection type of probe shall be used if build up is expected.

5.6 Fire and gas detection

5.6.1 General

TR1055 and TR2237, performance standards 3 and 7 state functional requirements related to fire and gas detection.

Fire and gas detectors may be of the smart addressable, e.g. field bus type.

5.6.2 Fire detection

5.6.2.1 General

Detector characteristics and calibration (pre-set sensitivity) shall ensure detection of a fire condition at an early stage. The detector shall be capable of operating under the conditions at the time that fire detection is needed.

Fire detectors should be self-monitoring.

Maximum response time of the fire detection function shall be evaluated and defined according to safety criticality. Response time for fire detection should not exceed 10 seconds.

5.6.2.2 Flame detectors

Where flame is predominating and according to flame wave radiation characteristic, flame detectors shall be of the multispectrum IR, or combination UV/IR, type.

The optical lens may have a built-in adjustable heater.

The sensitivity setting of the flame detector shall be determined according to distance from detector to fire object. Low sensitivity setting shall not be used.

5.6.2.3 Heat detectors

Heat detectors should be a combination of rate of rise, and fixed temperature.

Heat detectors should have local alarm indicators to visually indicate when detectors are in alarm mode

5.6.2.4 Smoke detectors

Optical smoke detectors should be used for detection of smouldering fire.

Ionic type, smoke detectors may be used where there is risk of energy intensive smoke generating fires, e.g., fire in high voltage switchgear caused by a short circuit.

Smoke detectors shall not be sensitive to water vapour.

Smoke detectors should have local alarm indicators to visually indicate when the detectors are in alarm mode.

Early warning sensors shall have higher sensitivity (0.4 % to 0.8 % light obscuring per metre) than the normal recommendation.

5.6.3 Gas detectors

IR gas detectors (HC) shall be the preferred technology whenever suited for the gas to be detected.

Gas detectors should be self-monitoring.

Catalytic detectors shall only be used if proper detection performance by other types is not achieved (e.g. on hydrogen gas).

H_2S detectors shall be used in areas with a risk of harmful H_2S concentrations, e.g. drilling mud/shale shaker area.

Ultrasonic gas leak detectors may be used. This type of detector may be beneficial in areas where the background noise level is constant.

Gas detector characteristics and sensitivity shall ensure that the presence of gas is not underestimated, e.g. gas concentration (point detectors), gas amount (optical beam detectors) or leakage rate (acoustic detectors).

The maximum response time of gas detection functions shall be evaluated and defined according to safety criticality, in order to ensure that total reaction time for each safety function can be fulfilled.

Typical response times that should be complied with unless faster responses are specified elsewhere, are as follows:

- IR detector response time (T90) should be less than 5 seconds for general area applications.
- Catalytic detector response time from low alarm (%LEL) to high alarm (%LEL) should be less than 10 seconds (considering test gas 50% LEL and T50(10s) and(T90 (30s))
- Acoustic detector response time including delays employed to improve false alarm immunity should not exceed 30 seconds.
- Upon gas exposure, the time from detector alarm limit is exceeded until alarm is presented/tagged on operator station should be less than 2 seconds.

Detectors shall give a fault alarm if the lens is contaminated or measurement path blocked.

5.7 Manual callpoint

Manual activation buttons shall be protected against inadvertent activation, e.g. by use of protective covers.

Manual callpoints shall have loop monitoring for open circuits/short circuits.

Push buttons shall be manually reset.

Each post shall be clearly marked, and a consistent colour coding shall be applied for the activation buttons.

5.8 Online analyser system

5.8.1 General

See TR0814 for use of online analysers for fiscal applications.

See API RP std 555 for recommended measurement principles.

Selection of analysers and design of analyser systems shall be carried out in close cooperation with the Company. Technology and components shall be field proven for the intended application and shall have documented performance under similar operating conditions.

Unless an inline analyser is being used, a sample probe should be used to take process samples.

Provisions shall be made for testing and calibrating each analyser. All necessary calibration/test equipment shall be supplied with the analyser. Where possible, the change over from process sample to test sample shall be controlled remotely from the LCR/CCR, by switching inline solenoid valves.

When designing an analyser system, the following requirements shall be evaluated:

- Analysis shall be performed on a representative sample (sampling in a turbulent flow regime);
- Results shall be within the required accuracy, response time and stability.
- Availability of system shall not be impaired by excessive maintenance and calibration requirements.
- The system shall be designed so that maintenance and calibration can be performed by one person.
- The analyser shall be provided with self-diagnostic facilities to give an alarm when it is not working properly or an error has occurred (fault alarm).

Analysers exposed to dirty processes (e.g. produced water) where sensors, measurement chamber, or sample system risk contamination shall have an automatic cleaning system.

Inline analysers shall be safely retractable during operation.

Manual sample points for laboratory analyses shall be provided at the online analyser sample point.

Where high availability of the analyser is important and there is a risk for deposits e.g. scaling or wax in operation, a duty-standby solution shall be evaluated. (e.g. for oil-in-water, O_2 for water injection).
Full duplication of analyser probes, sample lines, pressure reduction and gas conditioning systems shall be considered for critical applications.

Sample cylinders used in sour-gas service shall be either polytetrafluoroethylene (PTFE) coated or epoxy coated.

5.8.1.1 Sand detectors and erosion probes

As a minimum the acoustic sand detectors shall have possibilities for filtering of acoustic noise from other sources than the sand production. This will as a minimum require that the background noise is flat when the well or flow line is shut down.

Provision shall be made for online calibration of sand detectors via sand injection upstream of detectors.

Increasing erosion readings shall not be caused by mechanical failures or impacts.

5.8.2 Analyser House

Analyzer houses shall have interior gas detection.

The Analyser house should be large enough to allow sufficient headroom and free access to three sides of wall-mounted equipment.

A drain and vent system shall be provided for sample disposal unless the sample is returned to the process.

The back-pressure effect on analyzers connected directly to the system drain or vent shall be evaluated.

The location of sample supply and return connections shall be evaluated.

Adequate lighting shall be provided around the analyzers and other equipment.

Need for forced ventilation shall be considered.

The design shall be compliant to Statoil requirements in TR1055 and TR2237.

5.8.3 Sampling System

5.8.3.1 General

In order to monitor the true concentrations for complex or contaminated process waters, one must be able to control or minimise any phase separation during sampling and measuring stages.
To ensure that the sample under analysis is representative for the property to be measured and at conditions suited for the correct operation of the analyser, a careful study of the overall sampling system shall be carried out. The study shall include recommendations as specified by the analyser Supplier.

5.8.3.2 Sample Tapping

Samples that are taken from the process stream shall have the same composition as the average composition of the process over the whole cross section of the pipeline, at the location and time of sampling. Sampling/measuring in a true turbulent flow will give a representative analysis.

Placement of sample tapping points for hydrocarbon gas applications should follow the recommendations in ISO10715 Natural gas.

Placement of sample tapping points for liquid applications should follow the recommendations in ISO 3171 ch. 5.

For projects in North-America the requirements for sample tapping shall follow the recommendations in API MPMS Ch 14.1 for Natural Gas and API MPMS Ch 8.2 for liquid applications.

Inline analyzers shall be installed/positioned according to the same requirements as sample probes.

An ASTM type nozzle shall be used for obtaining steam samples and may be used for liquids. Steam samples shall be done via a cooler.

The “typical gas sample probe”, also referred to as a “straight tube sample probe”, can be used for gases. Flow impact probes should be considered where considered adequate.

Gas lines with streams free of entrained liquids and at flow conditions well above their dewpoint temperatures may be sampled with any probe design.

Lines that are operating at or near the gas stream dewpoint shall have a special probe designed to overcome the problems of condensation and liquid.

Where required, sample take-off & return probes shall be included. Typical installations use 2” flanged connections.

Each sample probe shall be fitted with a set of double block & bleed isolation valves.

The design of the probe shall take into account the possibility of resonant vibration being induced in the probe by high flow velocities in the pipeline.

Sampling devices shall not be installed in horizontal pipelines where contaminants are likely to be in suspension.

Mixing equipment shall be considered for installation upstream of the sampling probe, whenever required in order to achieve a representative sample.

5.8.3.3 Manual sample point for laboratory

Provision for manual sampling shall be accommodated, where required for back-up and calibration of online analysers. This manual sample point shall be located as close as possible to the sample point for the online analyser.

- All sample points should be installed together with a probe as specified in this document.
- All sample connections shall be fitted with a ball valve and a needle valve for tapping.
- Sample point shall be located for easy access.
- The sample excess shall be returned to the process, drain, vent or to the fuel system/flare.

5.8.3.4 Sample Transport Line

Representativity shall persist throughout the period of transfer of the batch. The sample shall be maintained in the same condition as at the point of extraction without loss of liquid, solids or gases and without contamination.

Sampling lines shall be as short as possible.
Sample volumes shall be kept to a minimum.

Time lag calculations shall be prepared and documented for each installation / stream. The transport time through the sample handling system shall be less than 2 minutes (i.e. to assure that a fresh sample is always analysed, to avoid "memory" effects).

The sample transport system shall be designed to convey the fluid to the analyser from the sample tap with the minimum of time lag (isokinetic flow).

Tubing and fittings used in the system shall not absorb components from the gas and thereby affecting the analysis result. Chemical leakproofing compounds (e.g. Loctite) which may emit components to the gas shall not be used. Tubing and fittings used in systems for measurement of H_2S shall be chemically treated (e.g. silicated) in order to avoid absorbsion or emission of H_2S. Tubing and fittings used for carrier gas shall also be chemically treated (e.g. silicated) in order to avoid affecting the measurement.

Fast loops systems are preferred due to small line lags and to a minimum of product waste.

Double piping from the probe to the analyser shall be considered for cleaning purposes when the sample is expected to be contaminated. If the sample contains slurries, then a cleaning facility downstream shall be considered. The connections and change over valves shall be primarily mounted on ground level or be accessible from accessible platforms.

Transmitters for flow, pressure and temperature and check-valves shall be used as necessary, to ensure that sample flow is adequate (isokinetic) and in the proper direction.

5.8.3.5 Sample Conditioning

It is imperative that the sample being analysed is suitable for the analyser. For example samples normally require some form of conditioning such as filtering, pressure control, flow control, temperature control, phase control or phase change prior to passing through the analyser.

As a general rule, all sampling systems shall contain suitable filters. Self-cleaning bypass type filters shall be used.

Conditioning may be required for the removal of dirt particles, liquid/gas entrainment, corrosive agents, etc. Special consideration shall be given to vapour samples containing a large proportion of steam or other components which needs to be condensed and removed before analysis.

When Analyser Houses are employed the sample conditioning equipment shall whenever possible be located on the outside walls of the analyser house and shall be adequately protected.

5.8.4 Online analyser used for authority reporting

Non-fiscal online analysers may be used for authority reporting. In such cases the analysers shall be subject to a technology qualification.

Analysers used for reporting of oil-in-water discharge to sea shall in each case be qualified according to Company requirements.

6 Final control elements

6.1 On/off valves

For technical requirements regarding actuators for on/off valves and accessories reference is made to TR0038 .

For requirements to montoring of safety critical valves reference is made to TR3138.

The pneumatic/hydraulic/mechanical arrangement of the ESV shall be designed and built so that the ESD action is never prevented, i.e. to bring the ESV to the safe state.
The PSD related arrangement of the ESV shall be designed and built such as to never prevent the ESD action, which is to bring the ESV to the safe state.

All on/off valves shall generally have CCR and local end position indication. All ESVs used for the sectionalisation purpose shall have local reset.

ESVs used in the blowdown system shall not have a local manual reset.

6.2 Control valves

6.2.1 General

TR2212 states the technical requirements for control valve and choke valves.

Globe valves should normally be used. However, other types (e.g. angle valves, ball valves, or disc/butterfly valves) may be used, subject to service conditions and application

Control valves shall be specified with leakage class III, class IV and class V. For applications with high differential pressure, shut off leakage class V shall be used.

Self-acting control valves shall be used only when a sufficient differential pressure exists.

6.3 Instrumentation

6.3.1 Air filter regulator

Air filter regulators shall be provided with an integral relief valve and micron filter. If regulator supply pressure is lost, downstream air pressure shall be relieved.

6.3.2 Positioner

In general, all control valves shall be equipped with a positioner, preferably an intelligent type.

The positioner shall preferably be installed as direct acting.

Valve positioners shall be provided with gauges to indicate supply pressure, control air signal, and positioner output pressure.

Electro-pneumatic valve positioners and pneumatic valve positioners with integral electro-pneumatic transducers shall not be used in a vibrating service conditions. The I/P transducer shall then be mounted separately from the valve and actuator assembly.

The total maximum inaccuracy of the signal conversion in I/Ps, or electro-pneumatic valve positioners, due to any limitation (e.g., repeatability, dead band, resolution, hysteresis, etc.), shall be less than 2.0%.

For intelligent positioners with 4-20 mA input and HART or Fieldbus communication and diagnostic possibilities, a documented baseline footprint/valve signature from the FAT shall be delivered electronically in a readable file format.

6.3.3 Limit switches

With the exception of North American facilities, limit switches shall be of inductive type complying with EN 60947-5-6 (NAMUR).

When limit switches are specified, a suitable junction box shall be mounted to the actuator or valve body for termination of flying leads.
Flying leads from limit switch to junction box shall be physically protected by design. If no other options are possible then AISI 316 SS tubing shall be used. When tubing is used, it shall be delivered with open ends for drainage.

Limit switches shall be activated by the movement of the valve shaft or stem and shall switch at less than 5% of the movement of the respective limit.

The limit switches shall normally be fitted inside a combined switch house/junction box, complete with terminals or be installed on rigid AISI 316 SS steel fixing plates or brackets and shall be easily adjustable.

Limit switches should preferably have integral LED indication.

6.3.4 Position Indicator

Both control valve and on/off valves shall have a local position indicator.
Position indicators on thermally insulated valves shall have possibilities for extension.

6.3.5 Solenoid valves and lock up devices

When used for control valves, the solenoid valve shall be installed in the pneumatic signal line between valve positioner and actuator.

Air lock-up devices shall be provided for all services requiring the control valve to remain in the position it had at loss of air. The lock-up device shall be mounted between positioner output and actuator.

When lock-up devices are used on control valves with solenoid valves, the solenoid valve shall be installed between lock-up valve and actuator.

6.4 Actuator requirements

6.4.1 Control valve actuator requirements

TR2212 states the technical requirements for control valve actuators.

6.4.2 On/Off valve actuator requirements

6.4.2.1 General

All actuators shall be designed and dimensioned to operate the valves under the worst case process conditions.

Actuator performance design shall be carried out with consideration of the overall valve response requirement.

The specified closing/opening time shall be independent of the process condition.

The safety margin stated for the valve application shall apply at any position of the valve stroke, and shall include the end position of the spring stroke for fail safe operations.

Actuators for safety valves shall generally be either electrical, pneumatic or hydraulic single acting with spring return. Where valve/actuator service exceeds the spring return design, hydraulic double acting actuator with accumulator shall be utilized.

The main selection criteria between these alternatives shall be LCC (Life Cycle Cost) including capital cost and normal maintenance costs, plus local availability of the type of energy supply. Operational wear, weight and dimensions shall also be considered, and shall be decisive for the selection where such factors are paramount. Access and serviceability shall also be considered for final selection between equally good alternatives.

Electric actuators shall be preferred for on-off valves without fail safe action, (PCS designation), if such actuation can be provided with the required speed of operation, and in accordance with safety restrictions. The nominal supply voltage shall be used for calculating torque/thrust on electrical actuators.

Double acting pneumatic actuators shall be avoided.

Pneumatic/hydraulic actuators shall be fitted with flushing/bleeding facilities.

6.4.2.2 Additional electrical actuator requirements

Electrical actuators used on safety valves shall either be well proven industrial types electrical actuators with spring return or electro-hydraulic double-acting.

UPS power supply should be standard for electrical actuated valves with safety action.

A three phase motor system should be selected to get a wider torque range and a more flexible and standard power configuration.

Diagnostic information from the actuator should be available on a serial communication link either wired or wireless. The condition monitoring system should be set up to handle the data from the actuator.

Operation of electrical actuated valves and the effects of transients and changes in power supplies during special operations like start-up of large equipment shall be evaluated.

6.4.2.3 Actuator Control Circuits

Type of circuit design shall be decided by the Purchaser. See TR0038.

The control circuits shall be designed for easy maintenance at location.

All accessories required for control of the actuator movement, such as hydraulic inlet filter, solenoid valves, manual valves, pressure gauges etc. shall be mounted on a control panel.

Valve control panels for on/off valves should be of the “compact” type, avoiding internal tubing and fittings.

Control panels for the different circuit types shall be standardised in size, layout and mounting arrangement. Valve control panels for on/off valves shall preferably be mounted on separate stands and not directly on actuator.

Devices for control of the speed in both directions shall be installed on the control unit. It shall not be possible to fully close the restrictors. The restrictors shall be easy to adjust during operation and shall have a fool proof locking device.

Pressure gauges shall indicate the internal pressures in the control unit.

Constant internal leakage shall not be designed into a control system, especially solenoid- and hydraulic/pneumatic valves.

Filter units to be used shall have a differential pressure indication.

The accumulators and their protection devices shall be assembled as a part of the control circuit panel, to form a composite unit, but they may also be required as independent units, depending on the space available in the plant. The vent outlets from bursting discs must be guided to a safe location.

Direct access for calibration and maintenance of all instruments shall be provided. It shall be possible to remove individual components without having to remove inter-connected components.

Hydraulic oil inlet and outlet ports shall be clearly marked "Oil Inlet" and "Oil Outlet", both on control panel and actuator.

Isolation valves for supply and return lines shall be installed as close to the instrument as possible.

Final layout arrangements proposed shall be confirmed and agreed by Purchaser upon fabrication.

Oil-mist lubrication systems shall not be used unless required for special applications (gear motors etc.).

A check valve shall be included on hydraulic circuit supply lines and pneumatic supply line.

For hydraulic operated valves with a check valve in the circuit supply line there will be a trapped-in liquid volume in the actuator/control circuit. In case there will be a temperature rise in this liquid volume (due to external heat input) the pressure may exceed allowable limits. To avoid this, a comined check/pressure relief valve should be included. A small volume nitrogen accumulator (nitrogen cushion separated from the oil by a diaphragm) can be used.

7 Additional information

7.1 Definitions and abbreviations

7.1.1 Definitions

Field instrumentation	Components installed in the field for detection, monitoring and control
Field instruments	Part of a system measuring physical conditions and transfer these into e.g. electrical signals that can be interpreted and used in SAS for safety, control and monitoring purposes, or local visualisation. E.g. instruments for pressure, temperature, level, and flow, fire and gas detectors.
Final control elements	Part of an instrumented system which implements the physical action necessary to achieve a predefined state. E.g. control valves, on/off valves, pumps and compressors (the latter two is part of mechanical competence area)
Commodity items	Components needed for transmitting field signals to/from instrumented systems, normally not tagged supplies. E.g. instrument tubing.
Gateway	Provides the connection to the host network. A WirelessHART gateway and the main host (PCDA) are interfaced using e.g. Modbus or Profibus, either over serial communication ports or over Ethernet. Configuration and administrative monitoring access is normally provided by http over Ethernet. The Gateway integrates both the Network and Security Manager into the same hardware.
Winterisation	Winterisation is the protection of instruments and instrument tubing against the effects of low ambient temperature; including hydration formation, condensation, viscosity changes and freezing.
Local Control Panel(LCP)	Local control panel to perform and monitor local actions on equipment like turbines or pumps
Instrument housing	Typically housing of a transmitter or an indicator
Instrument enclosure	A housing/cabinet to protect the field instrument from the ambient conditions.

7.1.2 Abbreviations

AC	Alternating Current
APS	Abandon Platform Shutdown
ATEX	ATmosphere EXplosive
BSP	BSPP Parallel & BSPT Tapered
CER	Central Equipment Room
CCR	Central Control Room
CPU	Central Processing Unit
DBB	Double Block and Bleed
DC	Direct Current
DP	Differential Pressure
ESD	Emergency ShutDown

ESV	Emergency Safety Valve
FAT	Factory Acceptance Test
F&G	Fire & Gas
FTC	Field Termination Cabinet
GoM	Gulf of Mexico
GRP	Glassfibre Reinforced Plastic
HART	Highway Addressable Remote Transducer
IP	Ingress Protection
IR	Infra Red
IS	Intrinsically Safe
LCI	Life Cycle Information
LCP	Local Control Panel
LCR	Local Control Room
LED	Light Emitting Diode
LER	Local Equipment Room
MCB	Miniature Circuit Breaker
MDS	Material Data Sheet
N/A	Not Applicable
NAS	National Aerospace Standard
NPT	National Pipe Taper threads
PCS	Process Control System
PSD	Process ShutDown
RIO	Remote Input Output
RTD	Resistance Temperature Detectors
SAS	Safety and Automation System
SIL	Safety Integrity Level
SS	Stainless Steel

7.2 Changes from previous version

1) General: Improved gramatics
2) Sect. 2.3.1 Clarified requirement for capacity of air supply
3) Sect. 2.3.2 Added requirement for flushing.
4) Sect. 3.1 Requirements for cable entry removed since this is covered by TR3023 and TR3024.
5) Sect. 3.2 Added requirements for alarm for reduced battery lifetime
6) Sect. 3.6.2 Added requirement for window in enclosures and process tubing connections for calibration and maintenance. Moved from ch.4
7) Sect. 3.9 Added reference for ex-requirement to fibre optic cables
8) Sect. 3.11.3 Added note to table 5 to clarify hardness of materials when using 316SS fittings with 6Mo tubing
9) Sect. 3.11.6 Requirements for junction boxes removed since this is covered by TR3023 and TR3024.
10) Sect. 3.13 Table 9; spare requirements for hydraulic distribution manifolds and electrical distribution removed
11) Sect. 3.14, Reference to TR2000 removed.
12) Sect. 3.14 Added requirements for packing ring when using parallel pipe thread connections
13) Sect. 3.15 Uncertainty requirements updated
14) Sect. 4.1 Added requirement for valve control panel to be mounted on separate stand
15) Sect. 4.1 Changed requirement for height of display elements.
16) Sect. 4.3 reference standard modified
17) Sect.4.4 Clarifying requirement to tubing clamps added
18) Sect. 4.4.2 New section, added requirements for cleaning of tubing
19) Sect 4.7 Added requirement for position of thermowell
20) Sect. 4.8 Added figure as an example 0 – 100% level
21) Sect 4.8 Added requirement for 0% reference point
22) Sect 4.9 added requirement for orifice plates

23) Sect. 4.9 Requirements in table 12 changed from shall to should
24) Sect. 4.10 Added requirement for pig tail in early warning detectors
25) Sect. 4.10 Added requirement for line of sight detectors
26) Sect 4.11 New section covering tagging and marking of field instruments
27) Sect. 5.2.1 Clamp-on replaced by non-intrusive
28) Sect 5.2.2 New version of standard ANSI/ASME PTC 19.3 added (wake frequency calculations)
29) Sect. 5.3.3 Added reference and additional requirement for restriction orifices.
30) Sect 5.3.4 Added reference
31) Sect. 5.5.2 Added requirement to clarify use of vibration level switch
32) Sect. 5.5.3 Removed requirement for forged steel and toughened steel in gauge glasses
33) Sect. 5.8.1.1 Added requirements to sand detectors and erosion probes
34) Sect. 6.1 Removed requirement for state monitoring and leak detection. New reference to TR3138.
35) Appendix A – Installation typicals added to document

7.3 References

Company requirements

- "Surface preparation and protective coating" (TR0042)
- "LCI Requirements" (TR2381)
- "Valve Actuators" (TR0038)
- "Metering systems for fiscal, allocation or custody transfer applications" (TR0814)
- "Performance Standards for Safety Systems and Barriers – Offshore" (TR1055)
- "HVAC Design and fabrication requirements" (TR1562)
- "Piping and Valve spec. (L-SP-200)" (TR2000)
- "Control & Choke Valves" (TR2212)
- "Safety design for Onshore Plants" (TR2237)
- "Valve Locking, Interlocking ans other Position Securing Systems" (TR2315)
- "Piping, Fabrication, Installation and Testing Specification" (TR2323)
- "Piping Detail Standard" (TR2325)
- "Process Safety" (TR3001)
- "Electrical system design, offshore units" (TR3021)
- "Electrical system design, onshore plants" (TR3022)
- "Electrical installation, offshore units" (TR3023)
- "Electrical installation, onshore plants" (TR3024)
- "Automation, Technical Requirements and Standards" (TR3030)
- "Automation technology" (TR3031)
- "Safety and Automation System" (TR3034)
- "Automation technology for equipment packages" (TR3131)
- "Testing and inspection of safety instrumented systems including safety related valves" (TR3138)
- "Vibration Protection and Diagnostic System" (TR3132)
- "Welding and Inspection of piping" (Norsok M-601)
- "Qualification of Manufacturers of special materials" (Norsok M-650)

Company guidelines

- "Field instruments" (GL3032 - hold)

International standards

- AGA Report no. 03, "Orifice Metering of Natural Gas and Other Related Hydrocarbon Fluids"
- AGA Report no. 07, "Measurement of Natural Gas by Turbine Meters"
- AGA Report no. 09, "Measurement of Gas by Multipath Ultrasonic Meters"
- AGA Report no. 11, "Measurement of Natural Gas by Coriolis Meter"

- ANSI/ISA 7.0.01, "Quality Standard for Instrument Air"
- ANSI/ASME "PTC 19.3 TW-2010"
- API/MPMS Ch 5.1, General Consideration for Measurement by Meter is missing from the references.
- API MPMS Ch 5.2, "Measurement of Liquid Hydrocarbons by Displacement Meters"
- API MPMS Ch 5.3, "Measurement of Liquid Hydrocarbons by Turbine Meters"
- API MPMS Ch 5.6, "Measurement of Liquid Hydrocarbons by Coriolis Meters"
- API/MPMS Ch 5.8, "Measurement of Liquid Hydrocarbons by Ultrasonic Flow Meters Using Transit Time Technology"
- API MPMS Ch 14.1, "Collecting and Handling of Natural Gas Samples for Custody Transfer"
- API MPMS Ch 8.2, "Standard Practice for Automatic Sampling of Liquid Petroleum and Petroleum Products
- API RP STD 555, "Process Analyzers"
- ASME B31.3 "Process Piping"
- ASME MFC-3M "Measurement of Fluid Flow in Pipes Using Orifice, Nozzle, and Venturi"
- ASTM A269 "Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service"
- ASTM E230 "Standard Specification and Tempertaure-Electromotive Force (EMF) Tables for Standardized Thermocouples"
- ASTM E1137 "Standard Specification for Industrial Platinum Reistance Thermometers"
- EN 60947-5-6, "Low-voltage switchgear and controlgear part 5-6: Control circuit devices and switching elements DC interface for proximity sensors and switching amplifiers (NAMUR)"
- IEC 60079, "Electrical apparatus for explosive gas atmospheres"
- IEC 60529, "Degrees of protection provided by enclosures (IP code)"
- EN 60751, "Industrial Platinum Resistance Thermometer sensors"
- IEC 61000, "Electromagnetic compatibility (EMC)"
- IEC 61158, "Industrial communication networks - Fieldbus specifications"
- IEC 61508, "Functional safety of electrical/electronic/programmable electronic safety-related systems"
- IEC 61511, "Functional safety - Safety instrumented systems for the process industry sector"
- EN 60534, "Industrial process control valves"
- IEC 60584, "Thermocouples"
- IEEE 802.11a "54 Mbit/s, 5 GHz standard"
- IEEE 802.11b "Enhancements to 802.11 to support 5.5 and 11 Mbit/s"
- IEEE 802.11g "54 Mbit/s, 2.4 GHz standard (backwards compatible with b)"
- IEEE 802.11n "Higher throughput improvements using MIMO (multiple input, multiple output antennas)
- ISA MC96.1 "Temperature Measurement Thermocouples"
- ISA S84.01, "Application of Safety Instrumented Systems for the Process Industry"
- ISO 228-1, "Pipe threads where pressure tight joints are not made on the threads"
- ISO 1000, "SI units and recommendations for the use of their multiples and certain other units"
- ISO 3171, "Petroleum liquids - Automatic pipeline sampling"
- ISO 4406, "Hydraulic fluid power -- Fluids -- Method for coding the level of contamination by solid particles"
- ISO 5167, "Measurement of fluid flow by means of pressure differential devices inserted in cuircular cross-section conduits running full"
- ISO 9951, "Measurement of gas flow in closed conduits - Turbine meters"
- ISO 10715, "Natural Gas - Sampling Guidelines"
- ISO 10790
- ISO15156/ NACE MR0175, "Materials for use in H_2S-containing environments in oil and gas production"
- NAS 1638 "Cleanliness Requirements fo Parts Used in Hydraulic Systems"
- NEMA "Ratings and IP Equivalency Chart"

- NEMA 250 “Ratings and IP Equivalency ChartEnclosures for Electrical Equipment (1000 Volts Maximum)”
- WirelessHART
- ISA100.11a

App A Installation typicals

Typical installation drawings below are for design guidance only. Principles shall however apply. Detailed hook-up drawings shall be issued for each project. Requirements for installation drawings are stated in TR2381.

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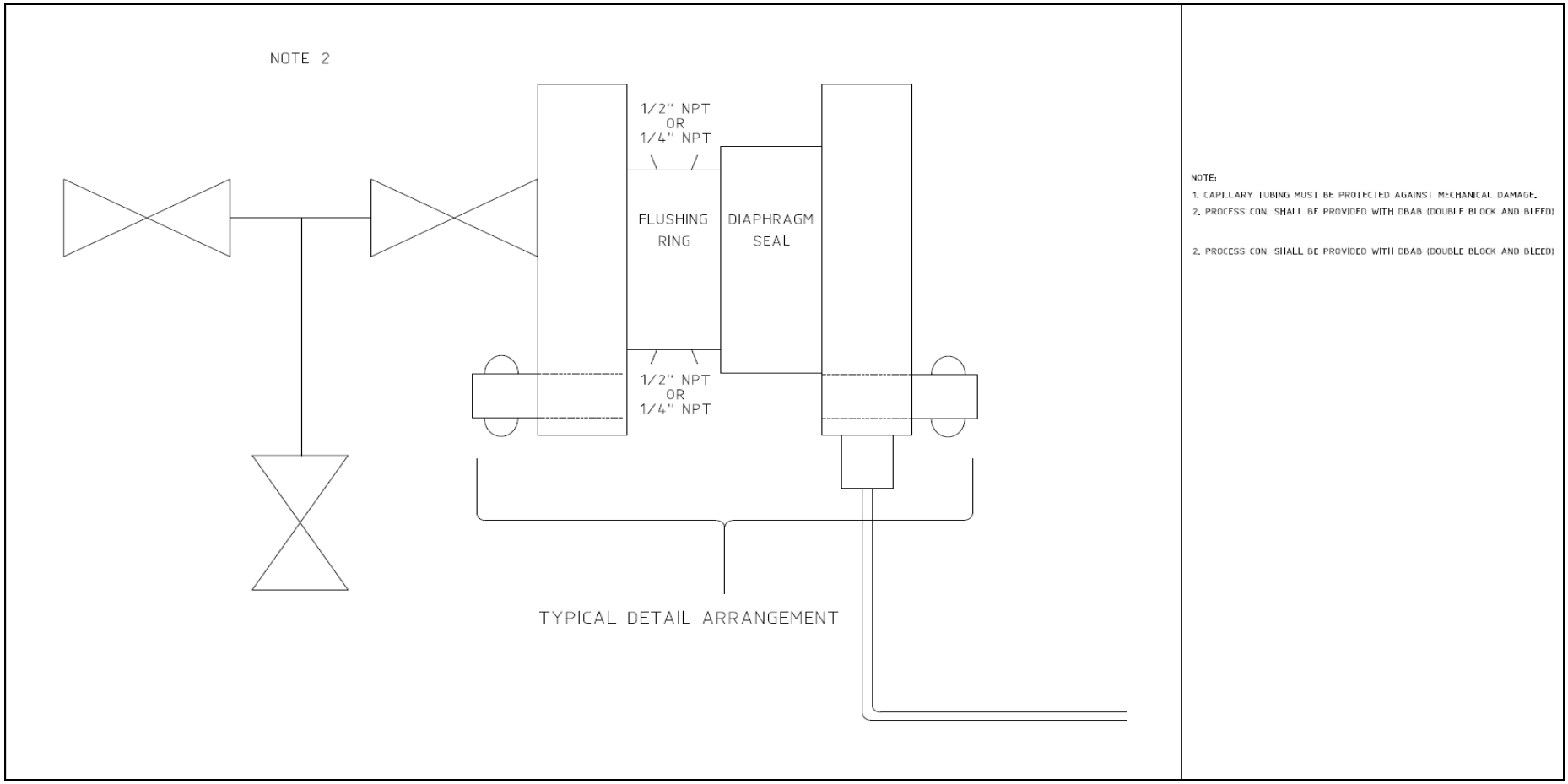
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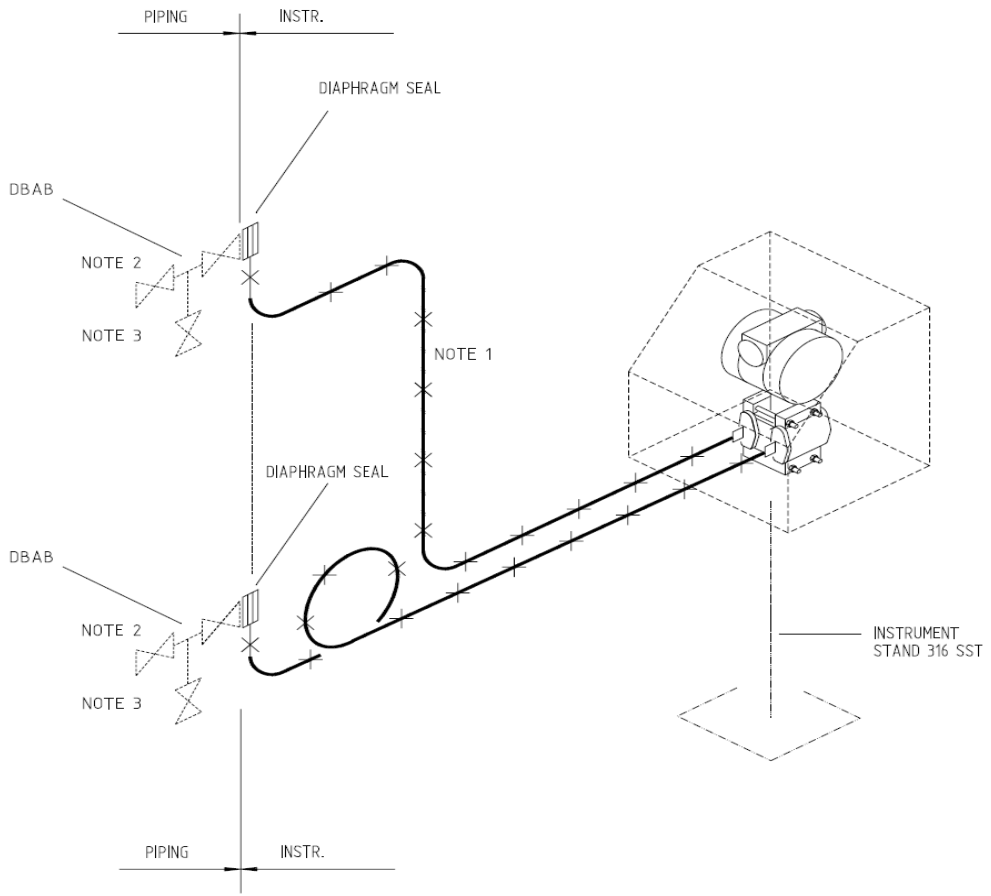
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A.1 Capillary and diaphragm seal arrangement



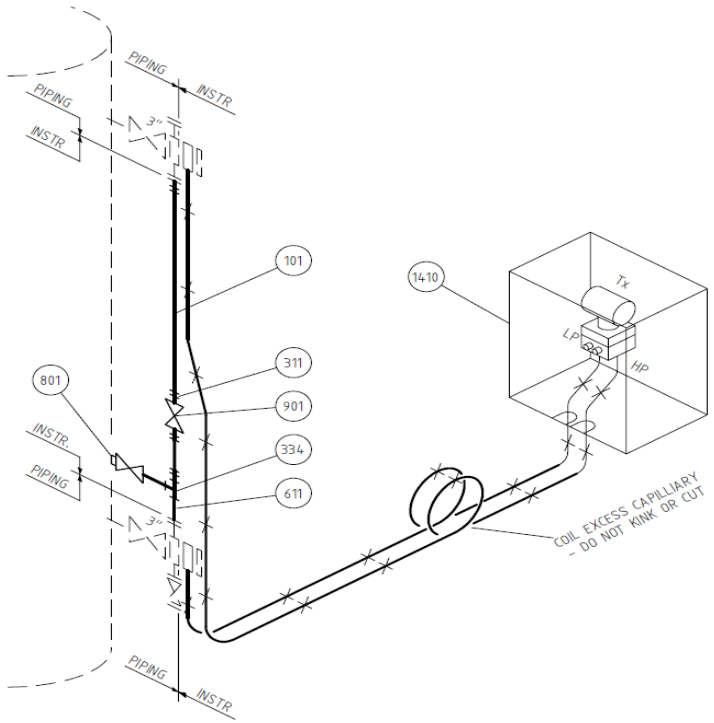
A.2 Differential pressure transmitter with capillary and diaphragm seal



NOTE:

1. CAPILLARY TUBING MUST BE PROTECTED AGAINST MECHANICAL DAMAGE.
2. PROCESS CON. SHALL BE PROVIDED WITH DBAB. (DOUBLE BLOCK AND BLEED)
3. INSTRUMENT CONNECTIONS TO BE 3" FLANGES.
4. RADIAL CONNECTION OF CAPILLARY IS PREFERED.

A.3 DP level transmitter, capillary type



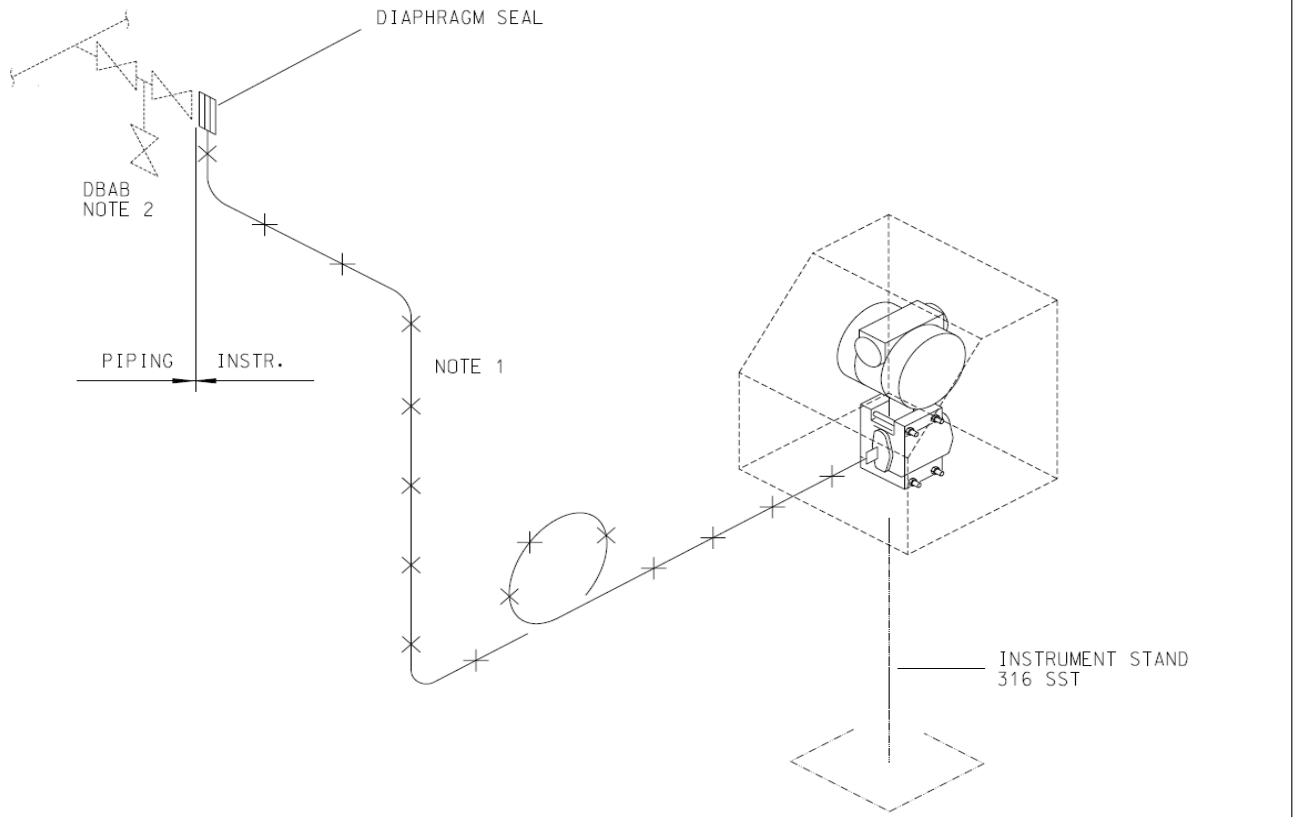
PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
101	TUBE	
311	MALE CONNECTOR	
334	TEE	
611	NIPPLE, HEX	
801	PLUG, HEX. HEAD	
901	BALL VALVE	
1410	INSTRUMENT ENCLOSURE – GRP/SS	

3.

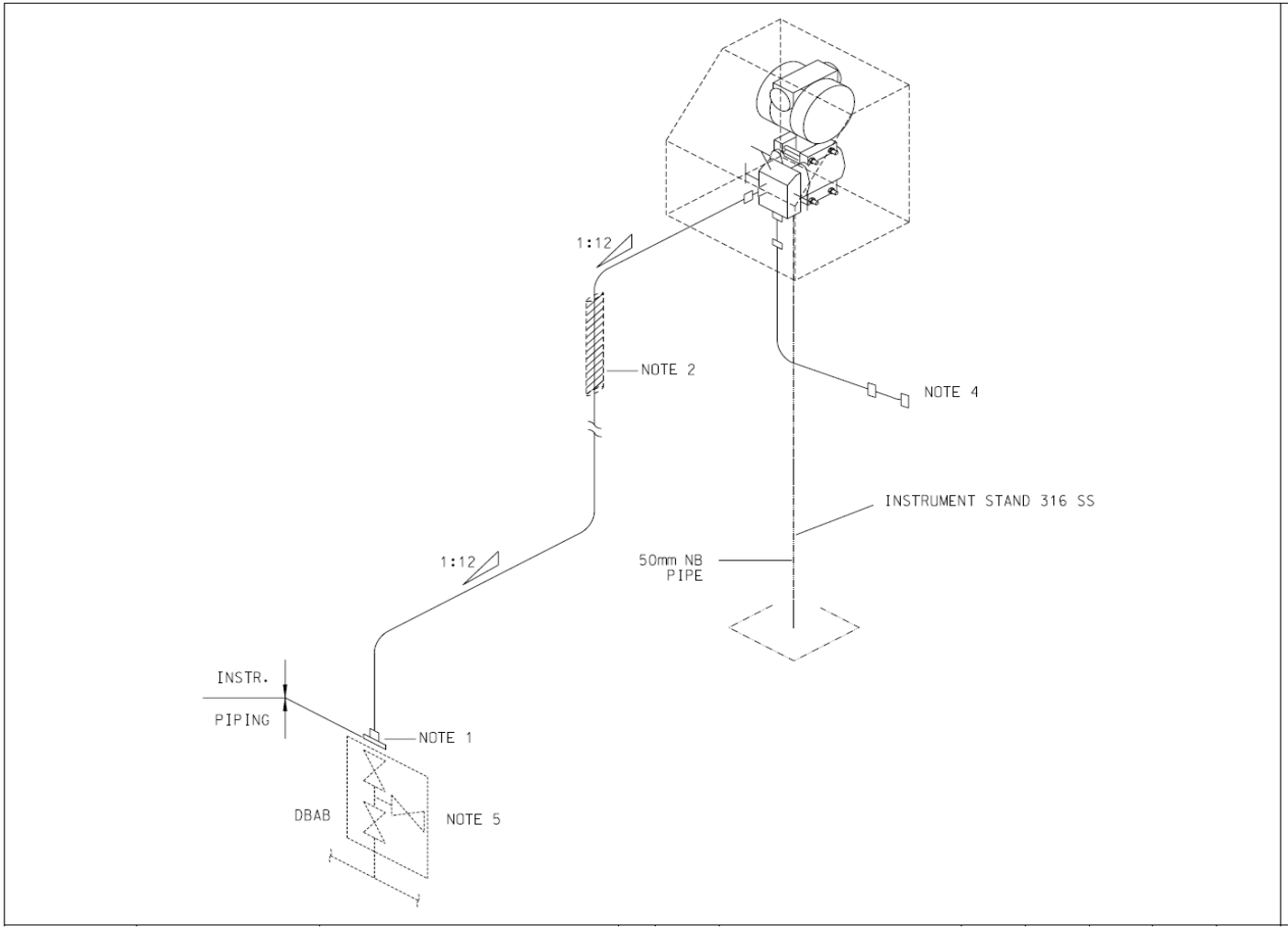
Product Spec. Sheet

A.4 Pressure transmitter with capillary and diaphragm seal



NOTE:
1. CAPILLARY TUBING MUST BE PROTECTED AGAINST MECHANICAL DAMAGE.
2. PROCESS CON. SHALL BE PROVIDED WITH DBAB (DOUBLE BLOCK AND BLEED)

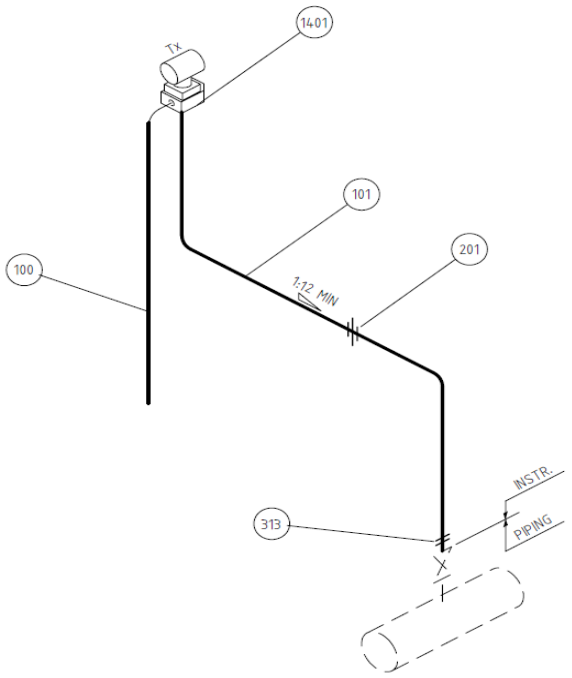
A.5 Pressure transmitter gas service



NOTE:

1. PROCESS CONNECTION
2. PRE INSULATED TUBING TO BE USED WHERE REQUIRED
3. WHERE OPERATING PRESSURE ARE ABOVE 5 BARG FOR LIQUID APPLICATION THIS HOOK-UP CAN BE USED
4. 6mm TEST CONNECTION WITH BLIND PLUGG
5. USE OF DBAB (DOUBLE BLOCK AND BLEED) IS SERVICE DEPENDANT

A.6 Pressure transmitter gas service

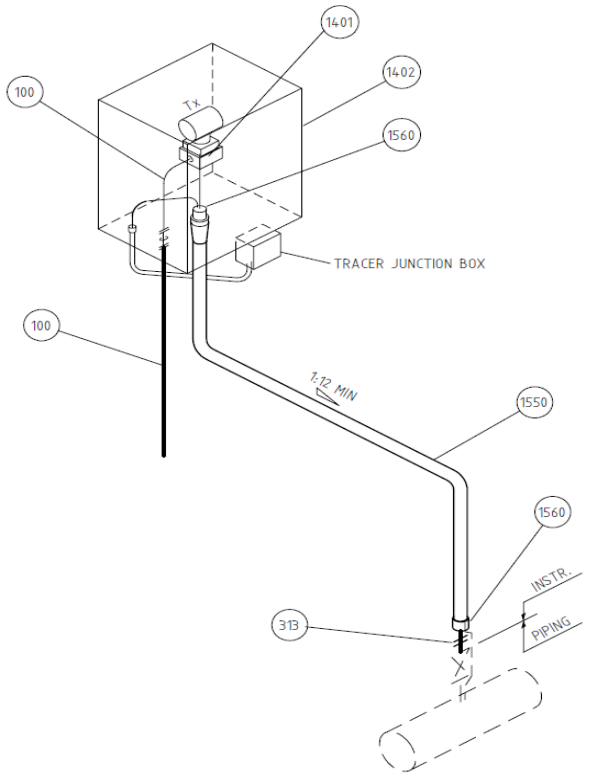


PRODUCT LIST

PRODUCT CODE		DESCRIPTION	REMARKS
100		TUBE	
101		TUBE	
201		UNION	
313		MALE CONNECTOR	
1401		2-way manifold	

Product Spec. Sheet

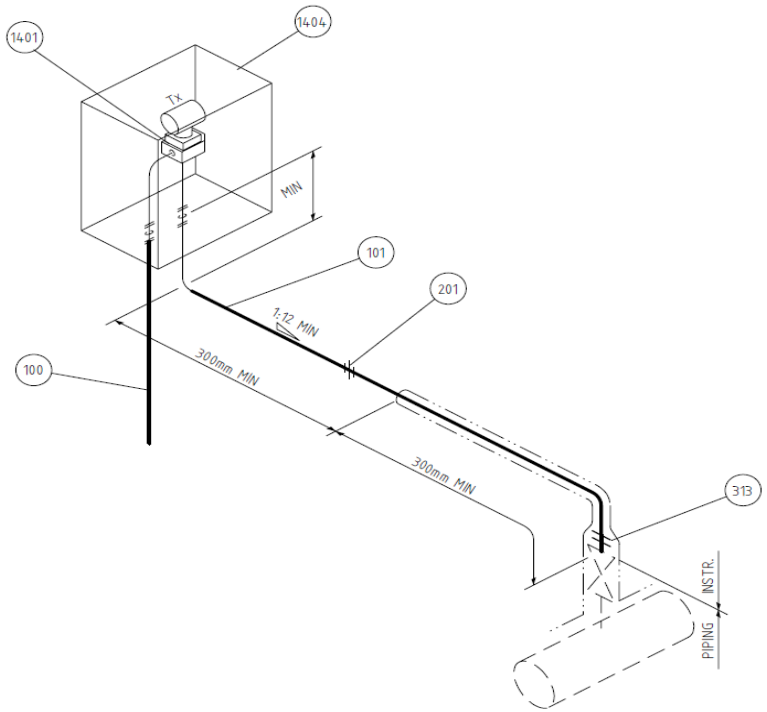
A.7 Pressure transmitter gas service heat traced



PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
100	TUBE	
313	MALE CONNECTOR	
1402	INSTRUMENT ENCLOSURE	
1550	PRE-INSULATED TRACED TUBES	
1560	PRE-INSULATED TRACED TUBE, SEAL END KIT	
1401	2-way manifold	

A.8 Pressure transmitter liquid/gas cryogenic

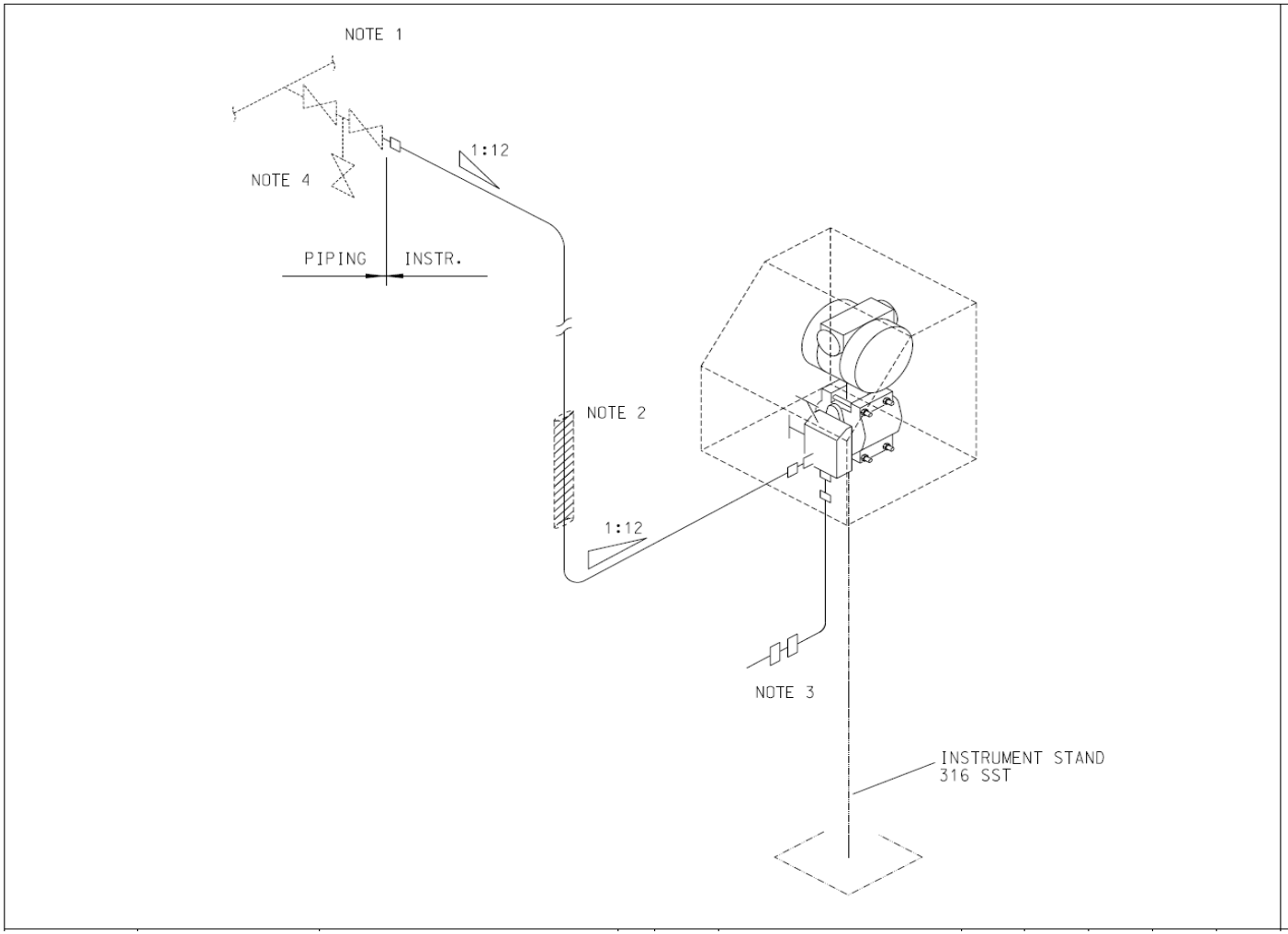


PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
100	TUBE	
101	TUBE	
201	UNION	
313	MALE CONNECTOR	
1404	INSTRUMENT ENCLOSURE GRP/SS	
1401	2-way manifold	

Product Spec. Sheet

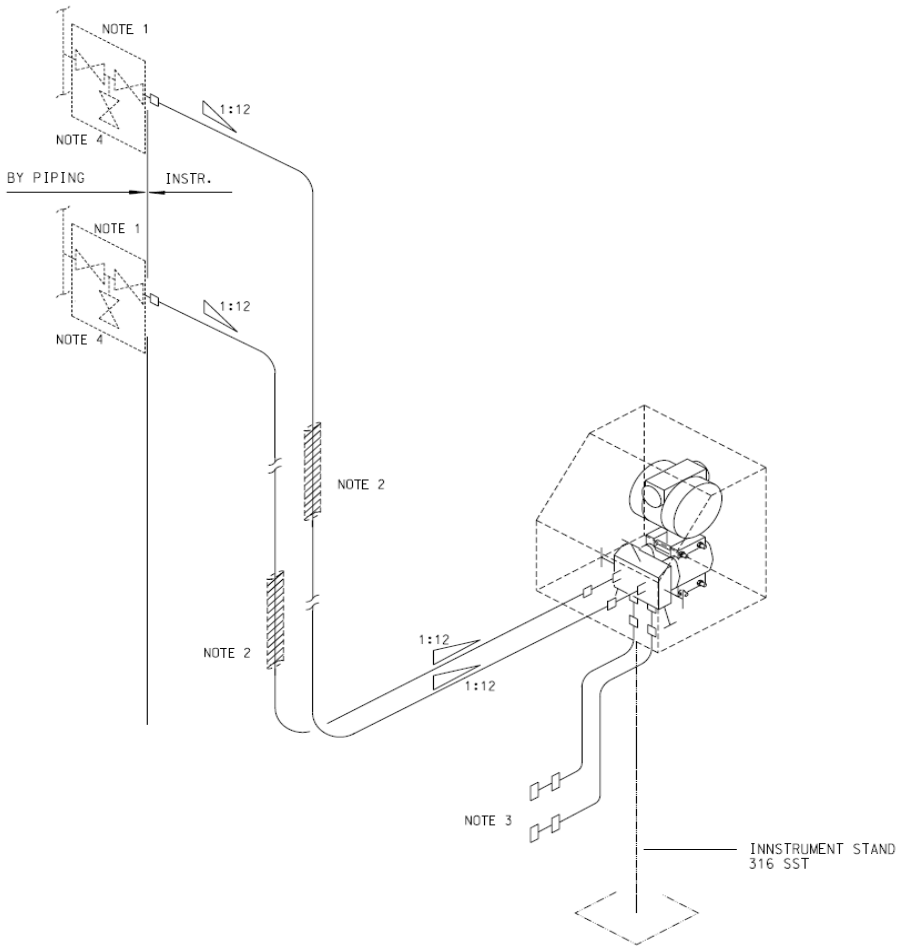
A.9 Pressure transmitter liquid service



NOTE:

1. PROCESS CONNECTION
2. PRE. INSULATED TUBING TO BE USED WHERE REQUIRED
3. 6mm TEST CONNECTION WITH BLIND PLUG
4. USE OF DBAB (DOUBLE BLOCK AND BLEED) IS SERVICE DEPENDANT

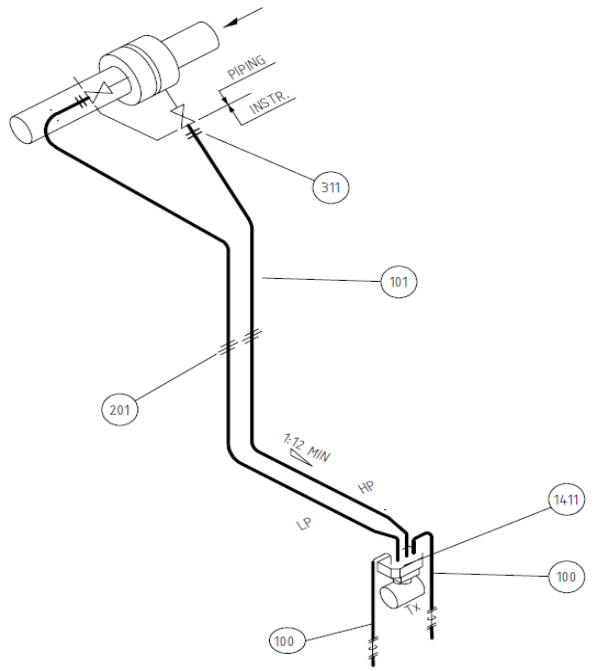
A.10 Differential pressure transmitter – liquid service



NOTE:

1. PROCESS CONNECTION
2. PRE-INSULATED TUBING TO BE USED WHERE REQUIRED
3. 6mm TEST CONNECTION WITH BLIND PLUG
4. USE OF DBAB (DOUBLE BLOCK AND BLEED) IS SERVICE DEPENDANT

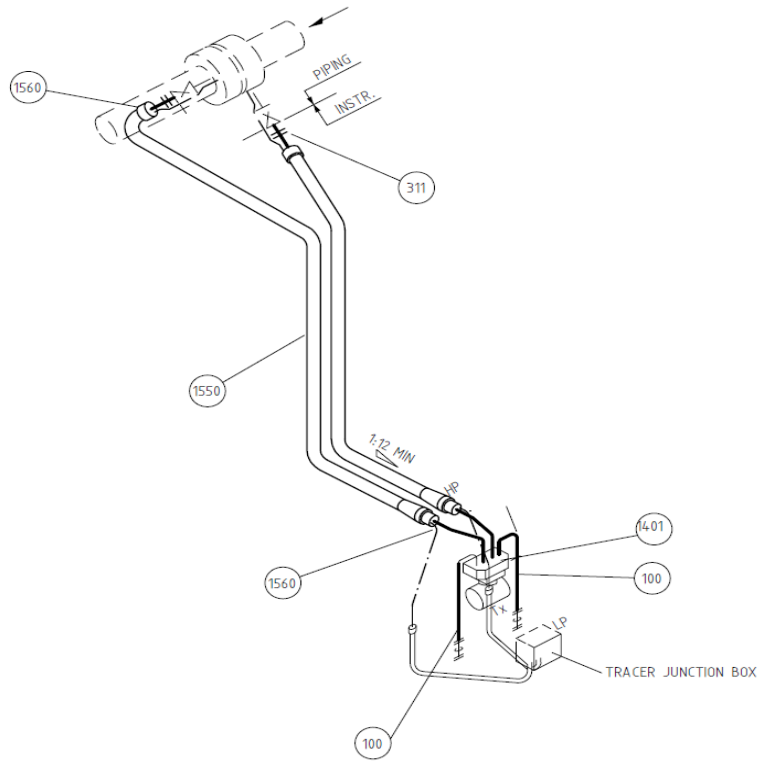
A.11 Differential pressure transmitter – liquid service



PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
100	TUBE, 6mm test connection with blind plug	
101	TUBE	
201	UNION	
311	MALE CONNECTOR	
1411	Manifold 5 valve	

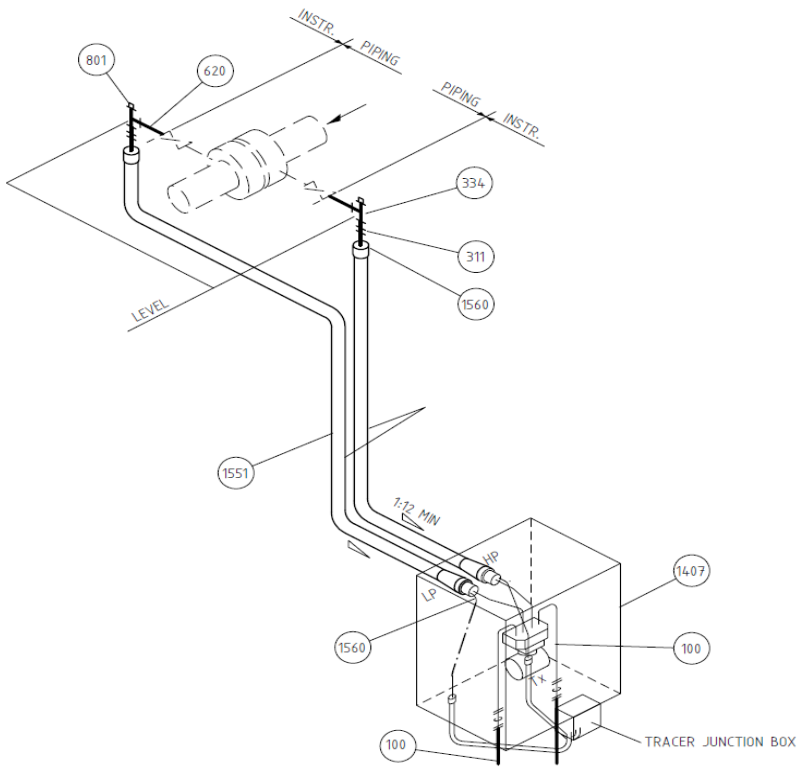
A.12 Differential pressure transmitter – liquid service heat traced



PRODUCT LIST

PRODUCT CODE		DESCRIPTION	REMARKS
100		TUBE, Test connection with blind plug	
311		MALE CONNECTOR	
1401		5-way manifold	
1550		PRE-INSULATED TRACED TUBE	
1560		PRE-INSULATED TRACED TUBE, SEAL END KIT	

A.13 Differential pressure transmitter – steam service

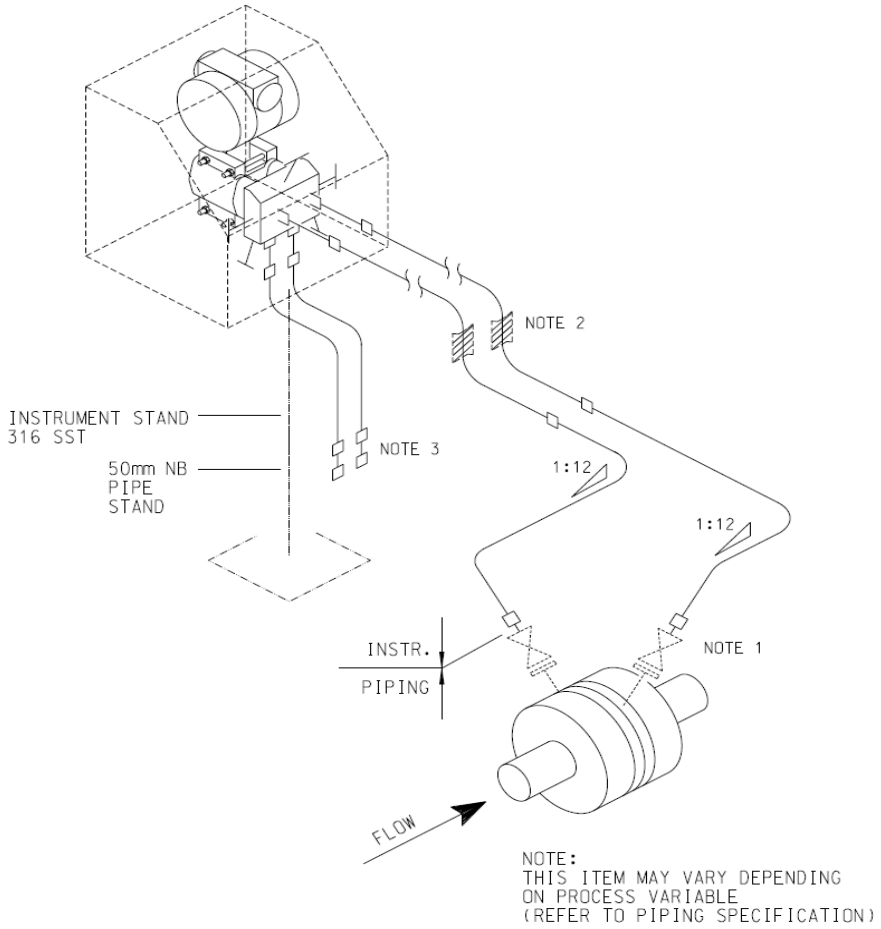


PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
100	TUBE, test connection with blind plug	
311	MALE CONNECTOR	
334	TEE, 1/2"NPT(F)	
620	PIPE NIPPLE, 1/2"NPT	
801	PLUG, HEX. HEAD, SCRD, 1/2"NPT	
1407	INSTRUMENT ENCLOSURE, GRP/SS	
1551	PRE-INSULATED TRACED TUBE	
1560	PRE-INSULATED TRACED TUBE, SEAL END KIT	

NOTES

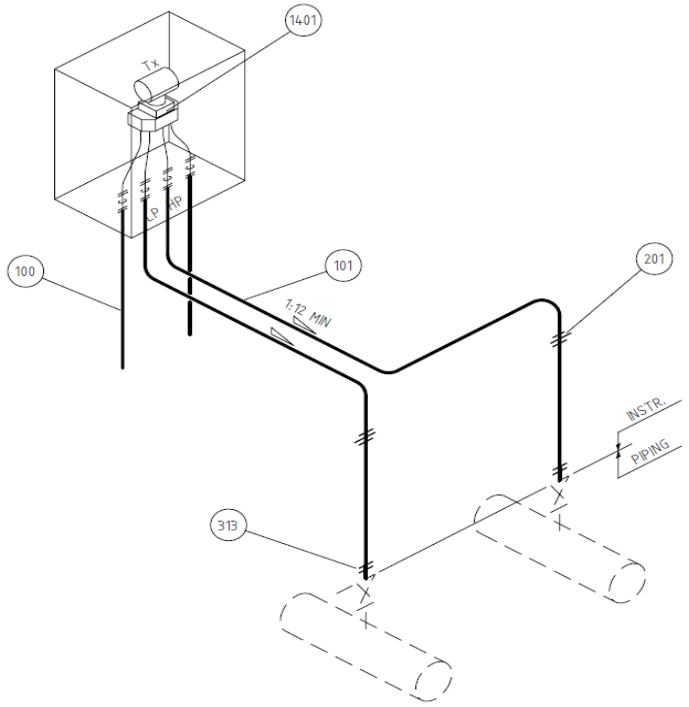
A.14 Differential pressure transmitter – gas service



NOTE:

1. PROCESS CONNECTION
   USE OF DBAB (DOUBLE BLOCK AND BLEED) IS SERVICE DEPENDANT
2. PRE-INSULATED TUBING TO BE USED WHERE REQUIRED
3. 6mm TEST CONNECTION WITH BLIND PLUG

A.15 Differential pressure transmitter – gas service

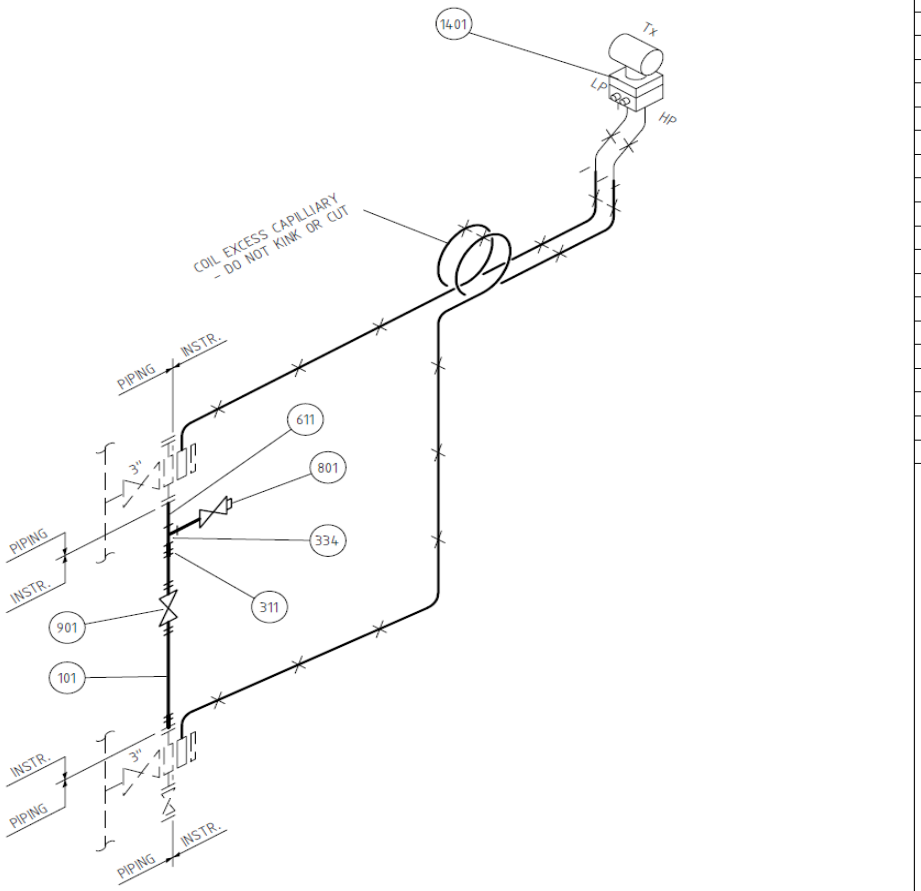


PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
100	TUBE	
101	TUBE	
201	UNION	
313	MALE CONNECTOR	
1401	5-way manifold	

Product Spec. Sheet

A.16 DP pressure transmitter capillary type

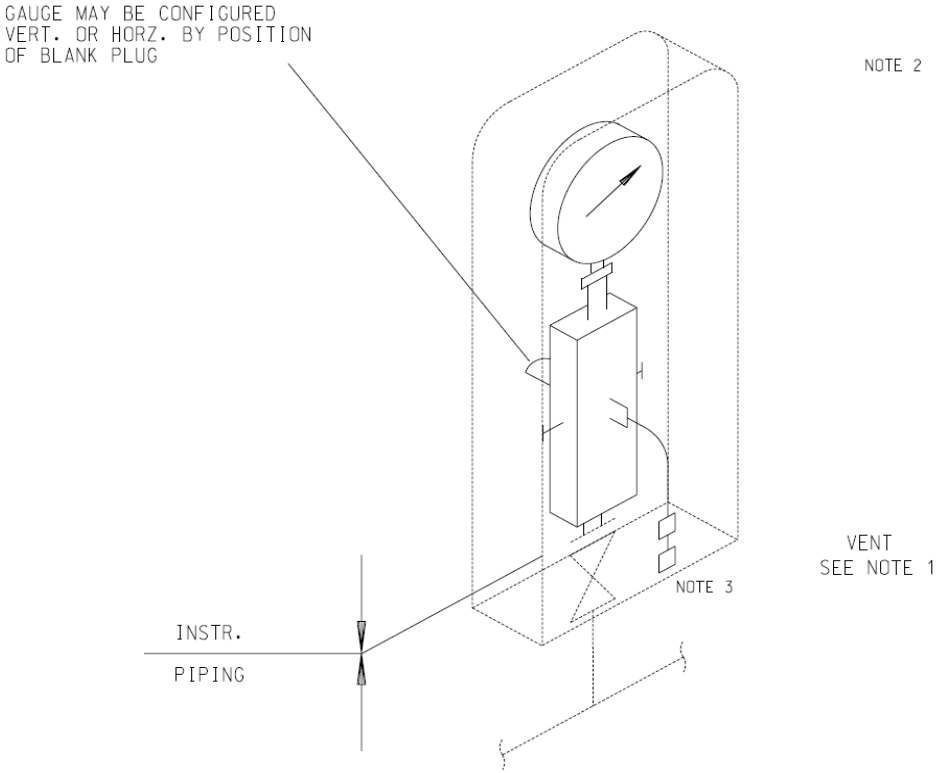


PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
101	TUBE	
311	MALE CONNECTOR	
334	TEE	
611	NIPPLE	
801	PLUG, HEX. HEAD	
901	BALL VALVE	
1401	5-way manifold	

Product Spec. Sheet

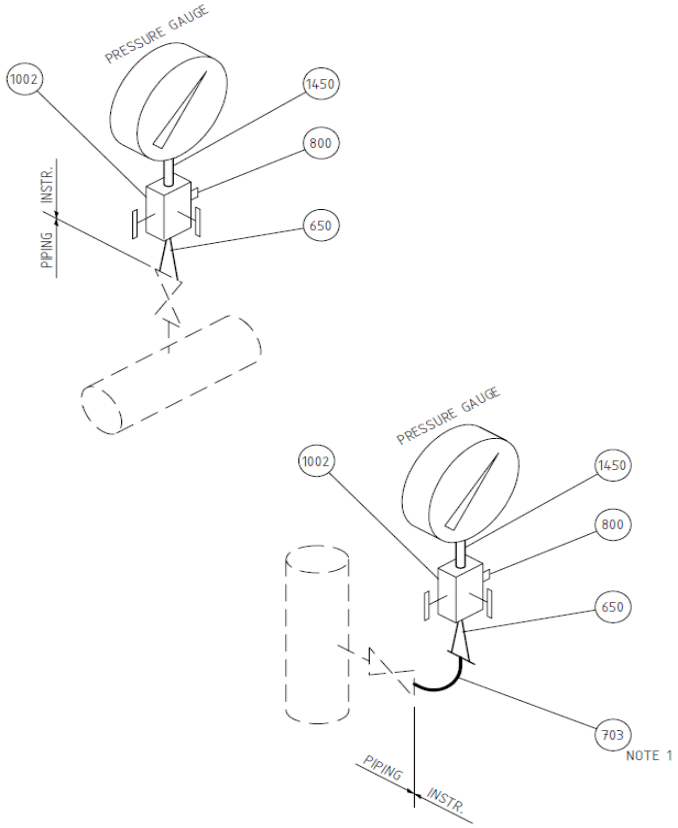
A.17 Pressure gauge



NOTE:

1. WHERE SPECIFIED SWIVEL VENT TO BE CONNECTED TO CLOSED DRAINS
2. GAUGE ADAPTER TO BE USED BETWEEN 3-WAY VALVE AND GAUGE
3. 6mm TEST CONNECTION WITH BLIND PLUG

A.18 Pressure gauge- gas service



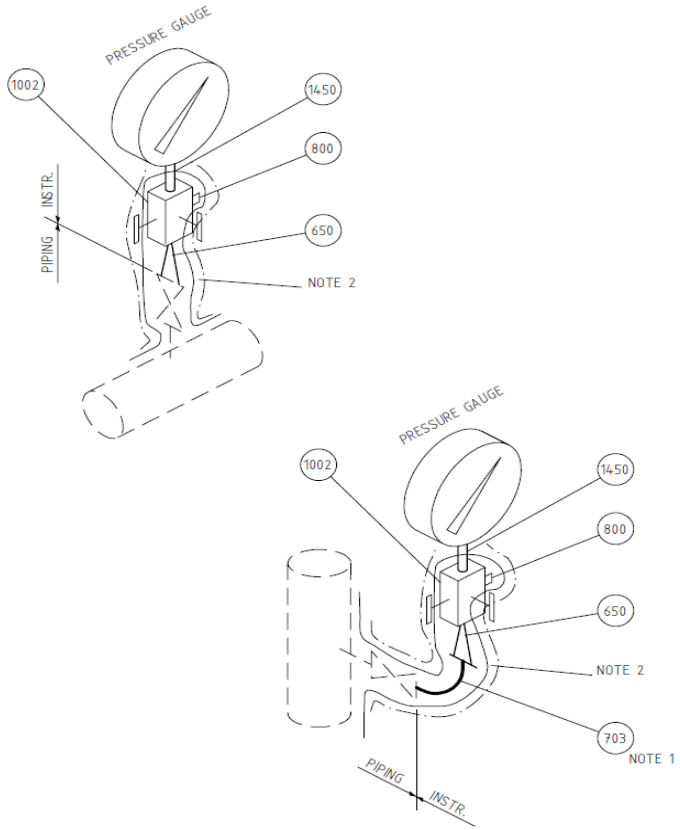
PRODUCT LIST

PRODUCT CODE		DESCRIPTION	REMARKS
650		NIPPLE	
703		STREET ELBOW	
800		PLUG, HEX. HEAD, SCRD	
1002		MANIFOLD, 2 VALVES	
1450		GAUGE ADAPTOR	

NOTES

1. ITEM 703 SHALL BE USED ON VERTICAL LINE INSTALLATIONS ONLY.

A.19 Pressure gauge – Gas heat traced



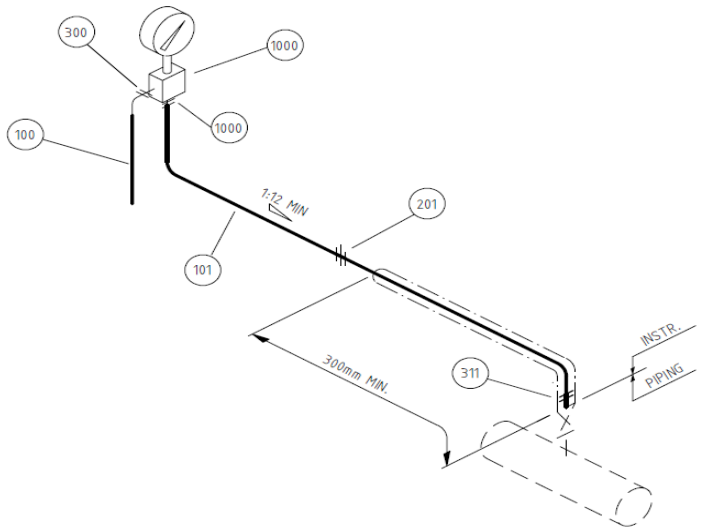
PRODUCT LIST

PRODUCT CODE	QTY	DESCRIPTION	REMARKS
650		SWAGE NIPPLE	
703		STREET ELBOW	
800		PLUG, HEX. HEAD, SCRD	
1002		MANIFOLD, 2 VALVES	SOFT SEAT VALVE
1450		GAUGE ADAPTOR, 1/2"NPT(M) x 1/2"NPT(F)	

NOTES

1. ITEM 703 SHALL BE USED ON VERTICAL LINE INSTALLATIONS ONLY.
2. ELECTRIC TRACER (BY OTHERS) TO BE INSULATED AS PER LINE REQUIREMENTS OR BY USING PROPRIETRY INSULATION TAPE.

A.20 Remote pressure gauge liquid/gas cryogenic

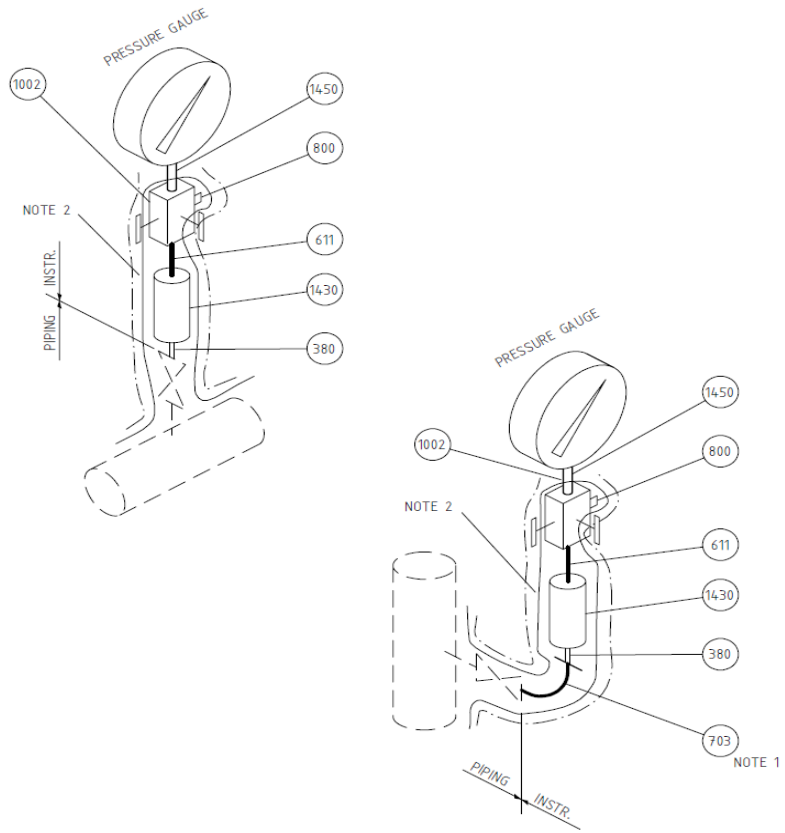


PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
100	TUBE	
101	TUBE	
201	UNION	
300	MALE CONNECTOR	
311	MALE CONNECTOR	
1000	MANIFOLD, 2 VALVES, 1/2" NPT, REMOTE MTD,	

NOTES

A.21 Pressure gauge – steam service



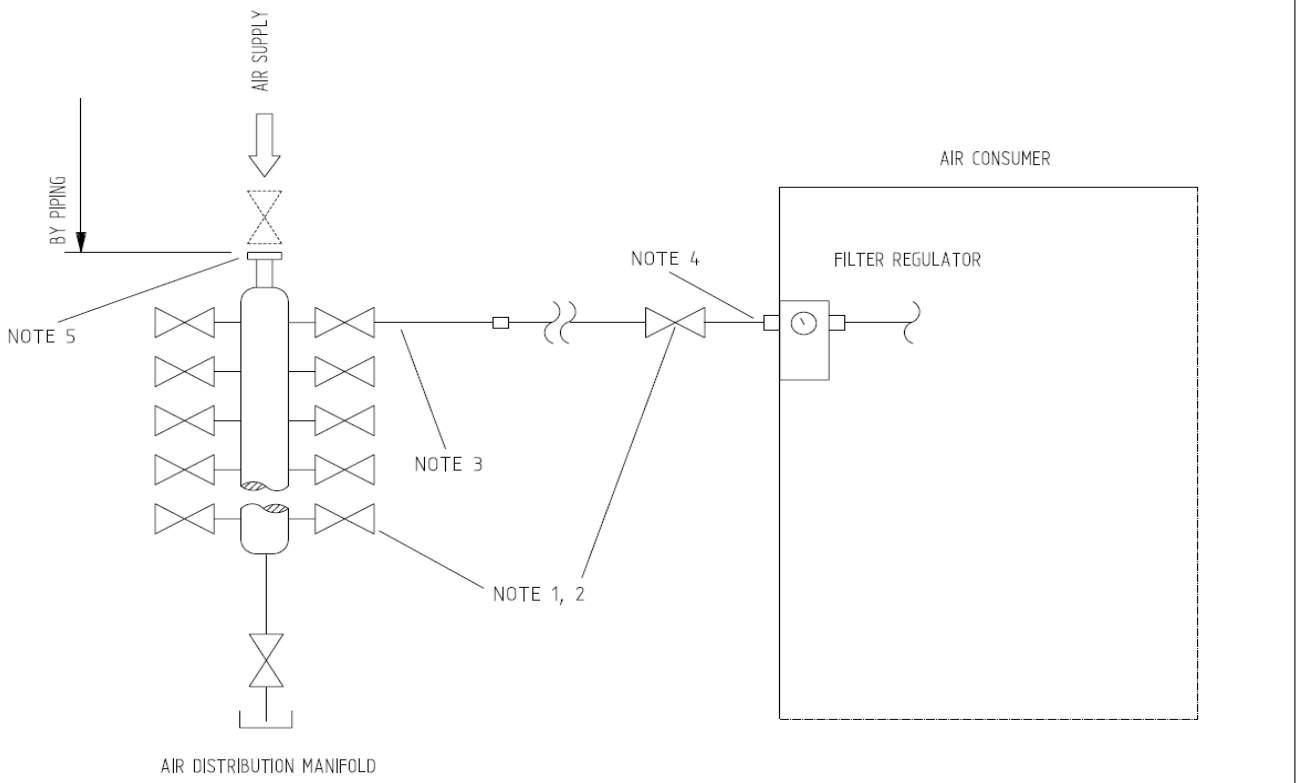
PRODUCT LIST

PRODUCT CODE	DESCRIPTION	REMARKS
380	REDUCING BUSHINGS	
611	NIPPLE, HEX.	
703	STREET ELBOW	
800	PLUG, HEX. HEAD, SCRD	
1002	MANIFOLD, 2 VALVES	
1430	SYPHON	
1450	GAUGE ADAPTOR	

NOTES

1. ITEM 703 TO BE USED ON VERTICAL LINE INSTALLATIONS ONLY.
2. ELECTRIC TRACER (BY OTHERS) TO BE INSULATED AS PER LINE REQUIREMENTS OR BY USING PROPRIETRY INSULATION TAPE.

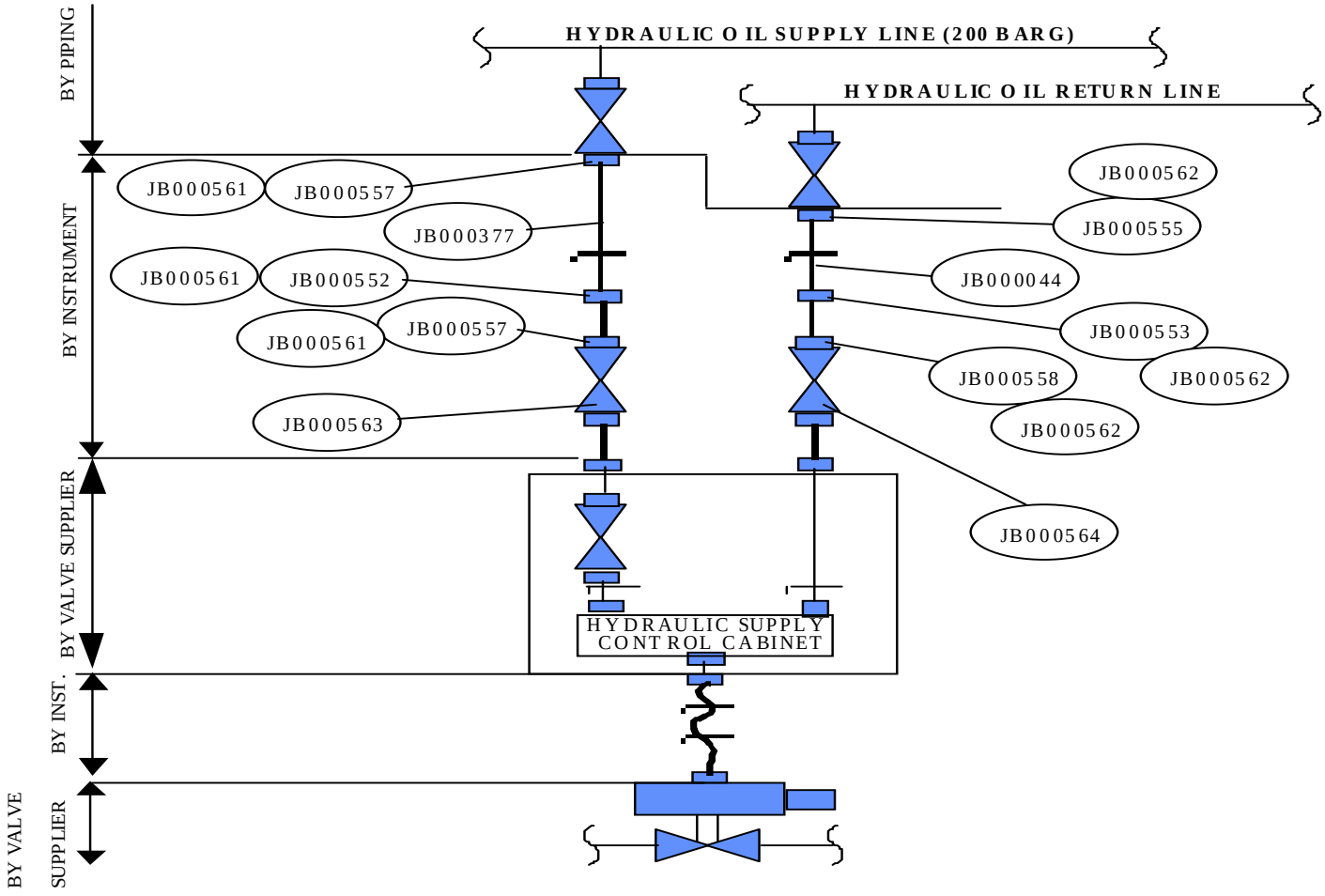
A.22 Air distribution of manifold w/consumer



NOTE:

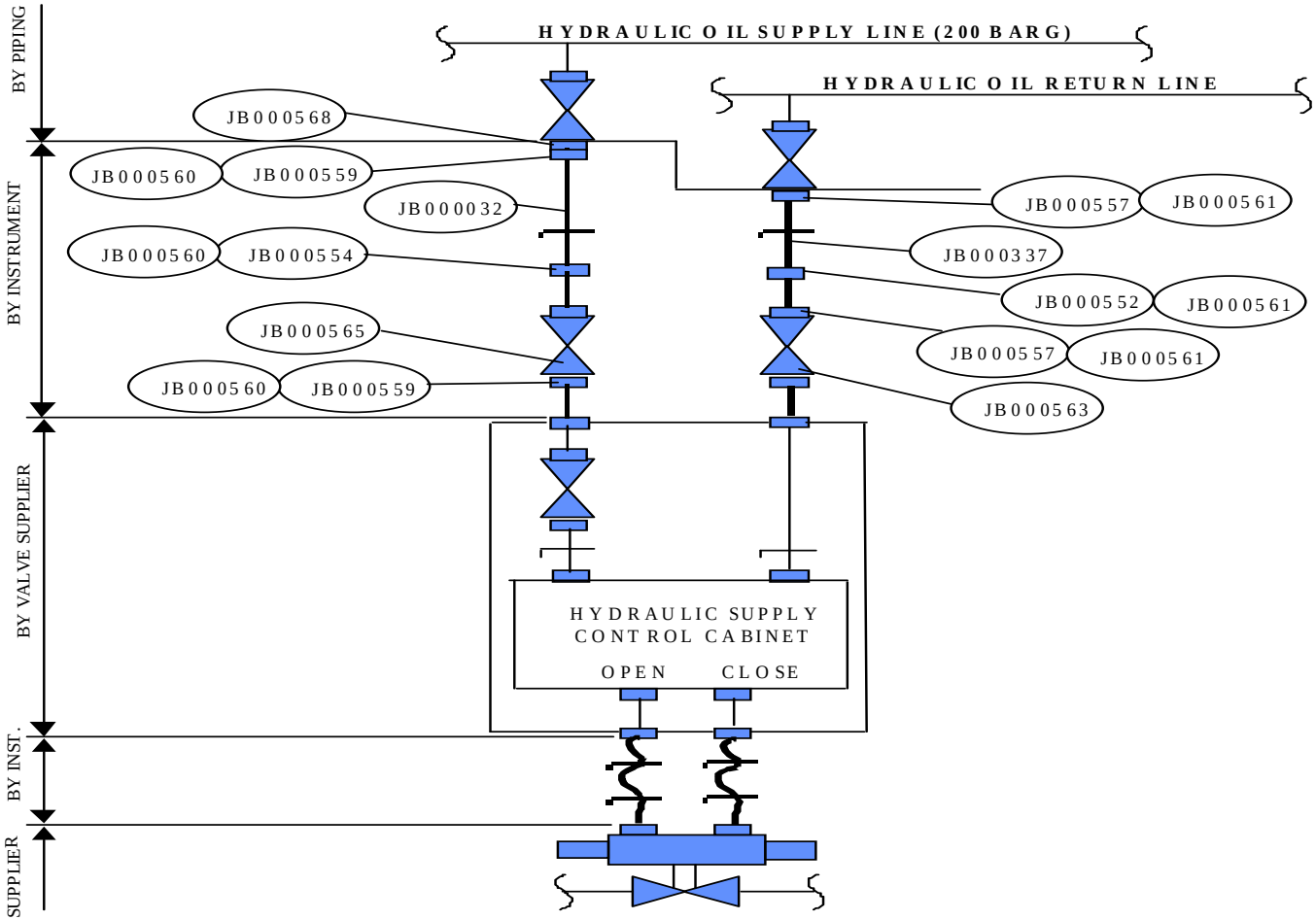
1. FULL BORE RISING PLUG VALVE, OR BALL VALVE WITH LOCKING DEVICE.
2. ALL SPARE VALVES SHALL BE PROVIDED WITH CAP.
3. ALL USERS SHALL HAVE TAG NUMBER ON TUBING.
4. MANIFOLD TAG TO BE PUT ON END.
5. IF TUBING IS USED FROM MAIN HEADER TO DISTRIBUTION MANIFOLD 25mm SHALL BE SELECTED.

A.23 Hydraulic distribution w/consumer



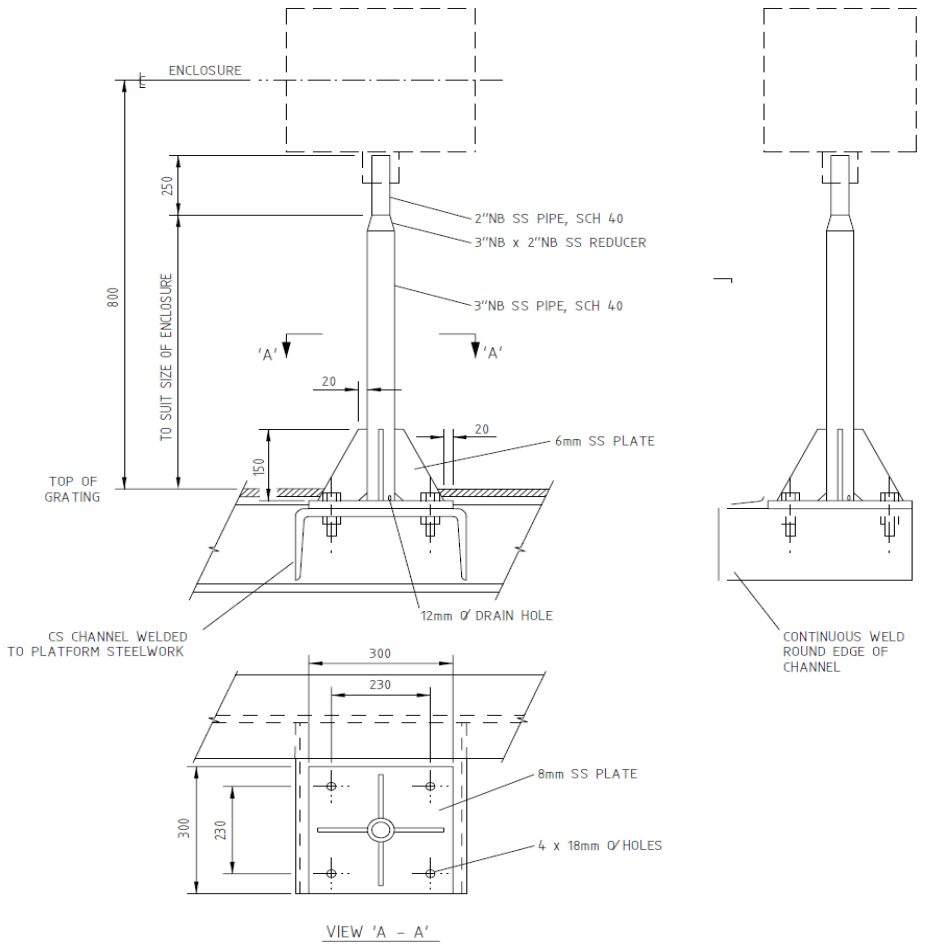
Item	Description	Qty
JB000552	TUBE UNION, 20mm OD, . G20S71X	1.0
JB000562	25mm FUNCTION NUT FM25S71	5.0
JB000561	20mm FUNCTION NUT FM20S71	5.0
JB000553	TUBE UNION, 25mm OD, G25S71X	1.0
JB000555	MALE CONN. 25mm OD X 3/4" BSPP , GE25SR3/4EDOMD71	1.0
JB000377	INSTRUMENT TUBE, IT-20 OD X 2.0MM 316SS.	12.0M
JB000557	MALE CONN. 20mm OD X 3/4" BSP , GE20SREDOMD71	3.0
JB000563	BALL VALVE 3/4" BSPP FEM. PISTER RKH4425 ¾" G	2.0
JB000564	BALL VALVE 1" BSPP , 316SS PISTER RKH4425 1" G	1.0
JB000044	INSTRUMENT TUBE, IT-25.OD X 2.0 MM 316SS.	12.0M
JB000558	MALE CONN. 25mm OD X 1" BSPP GE25SREDOMD71	2.0

A.24 Hydraulic distribution w/consumer



Item	Description	Qty
JB000032	INSTRUMENT TUBE, IT-10.0 X 2.0MM 316SS. 10.0mm OD X 2.0mm WT.	12.0M
JB000561	20mm FUNCTION NUT FM20S71	5.0
JB000560	10mm FUNCTION NUT FM10S71	5.0
JB000568	THREAD REDUCER/EXPANDER ¾"M x ½"F RI3/4EDX1/271	1.0
JB000554	TUBE UNION, 10mm OD, G10S71X	1.0
JB000552	TUBE UNION, 20mm OD, G20S71X	1.0
JB000559	MALE CONN. 10mm OD X 1/2" BSPP GE20SREDOMD71	3.0
JB000565	BALL VALVE ½" BSPP 7723F8Y/BSPP	1.0
JB000377	INSTRUMENT TUBE, IT-20mm OD X 2.0mm WT 316SS.	12.0M
JB000557	MALE CONN. 20mm OD X 3/4" BSPP GE20SRED71X	3.0
JB000563	BALL VALVE 3/4" BSPP RKH4425 ¾" G	1.0

A.25 Instrument support – housed instrument



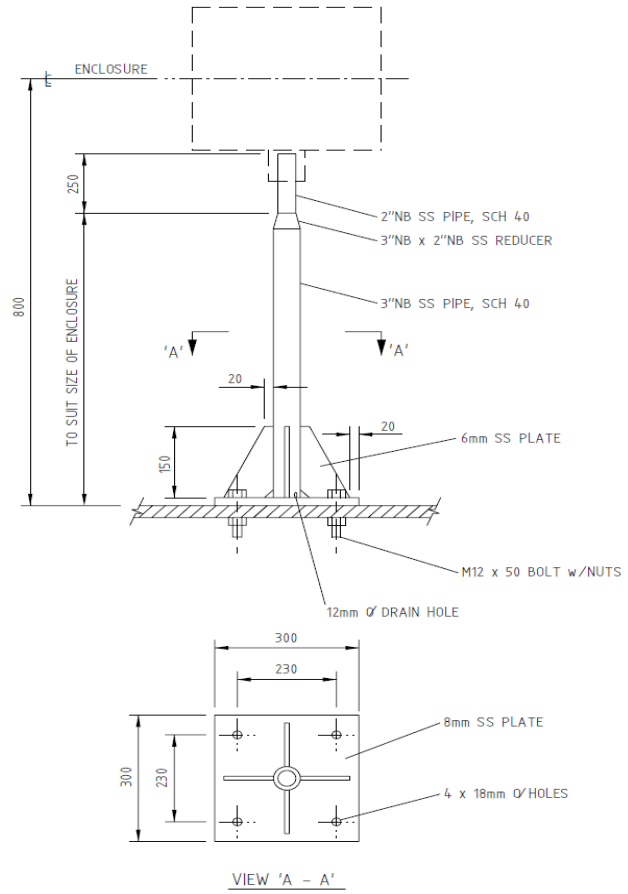
PRODUCT LIST

PRODUCT CODE	QUALITY CODE	QTY	DESCRIPTION	REMARKS

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETRES.

A.26 Instrument support – platform mounted



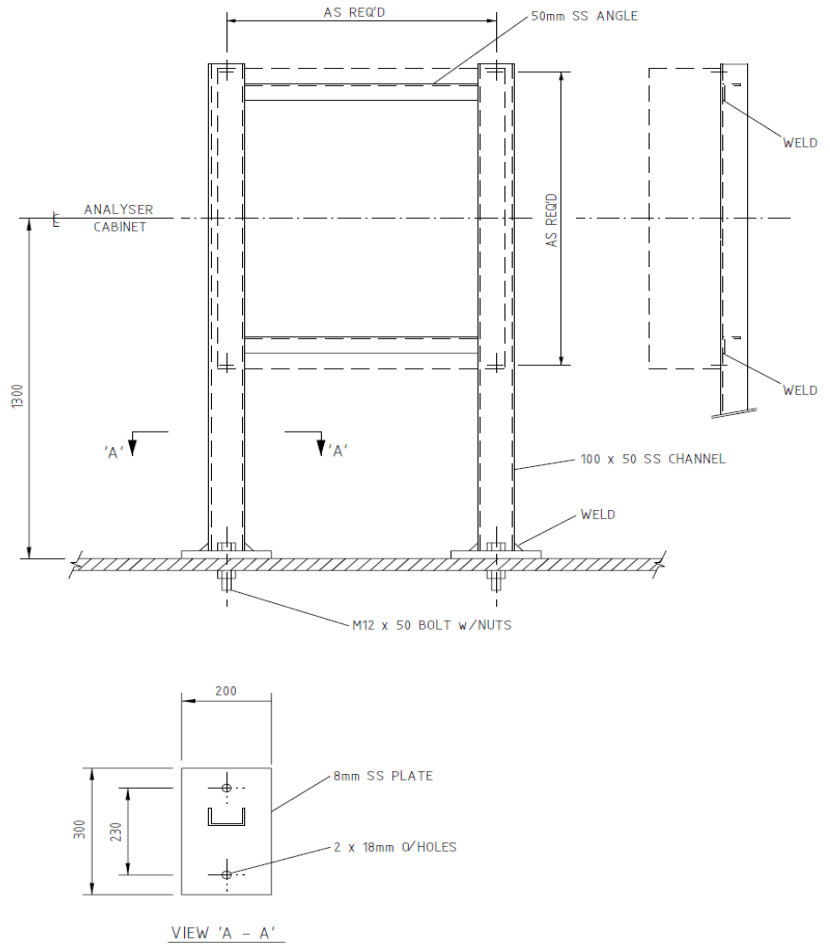
PRODUCT LIST

PRODUCT CODE	QUALITY CODE	QTY	DESCRIPTION	REMARKS

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETRES.

A.27 Instrument support system – local analyser system



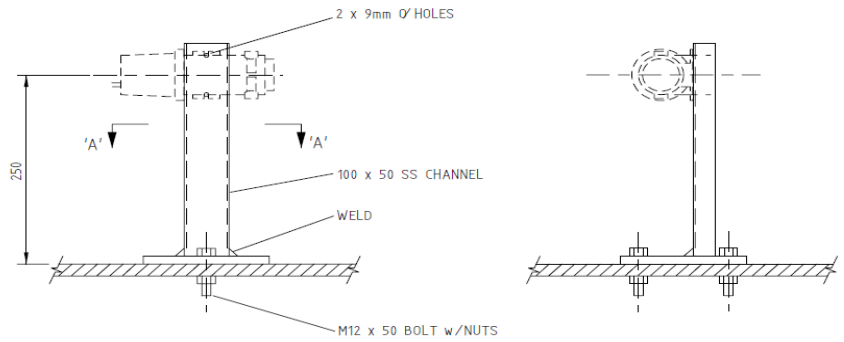
PRODUCT LIST

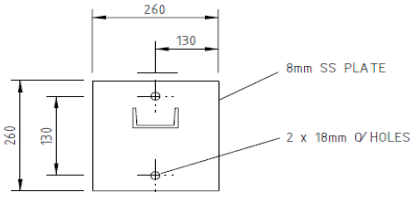
PRODUCT CODE	QUALITY CODE	QTY	DESCRIPTION	REMARKS

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETRES.

A.28 Instrument support – gas detector





VIEW 'A – A'

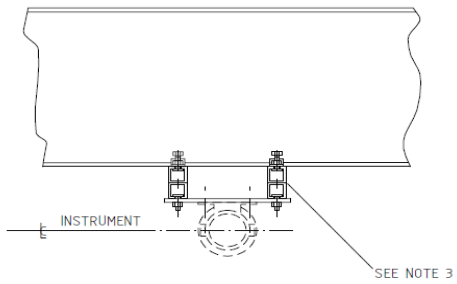
PRODUCT LIST

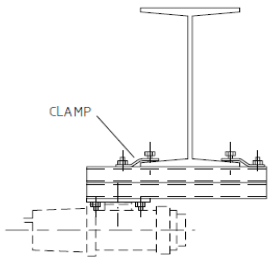
PRODUCT CODE	QUALITY CODE	QTY	DESCRIPTION	REMARKS

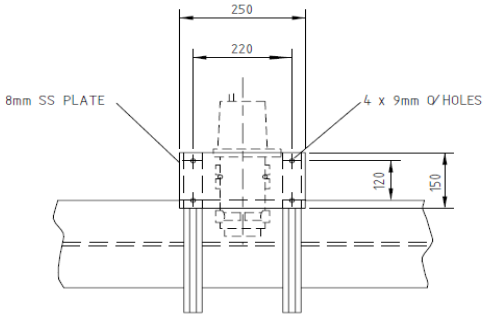
NOTES

1. ALL DIMENSIONS ARE IN MILLIMETRES.

A.29 Instrument support – gas detector







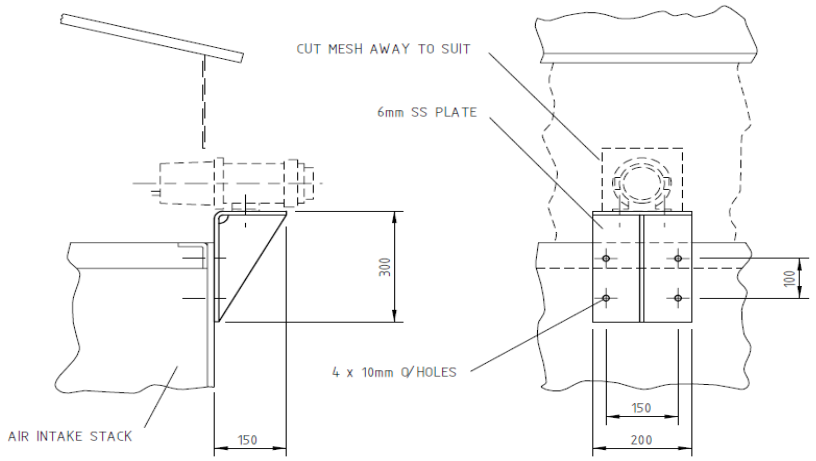
PRODUCT LIST

PRODUCT CODE	QUALITY CODE	QTY	DESCRIPTION	REMARKS

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETRES.

A.30 Instrument support – gas detector



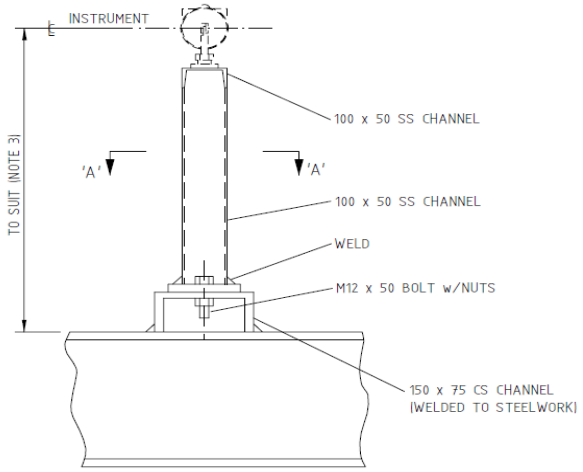
PRODUCT LIST

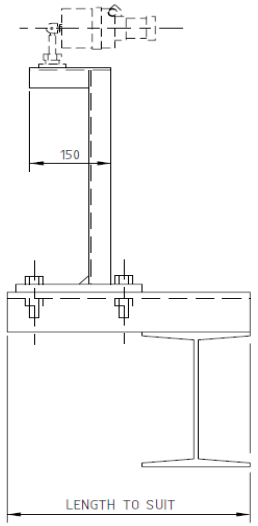
PRODUCT CODE	QUALITY CODE	QTY	DESCRIPTION	REMARKS

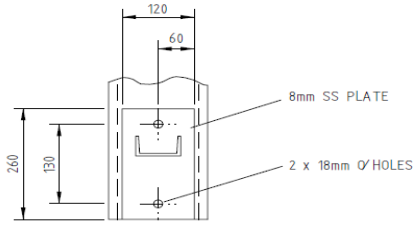
NOTES

1. ALL DIMENSIONS ARE IN MILLIMETRES.

A.31 Instrument support – flame detector







VIEW 'A – A'

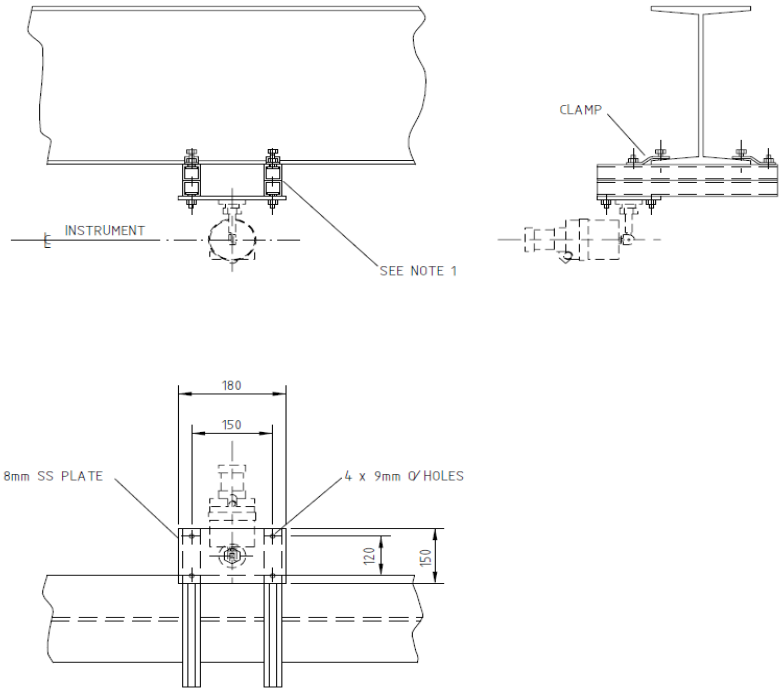
PRODUCT LIST

PRODUCT CODE	QUALITY CODE	QTY	DESCRIPTION	REMARKS

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETRES.

A.32 Instrument support – flame detector



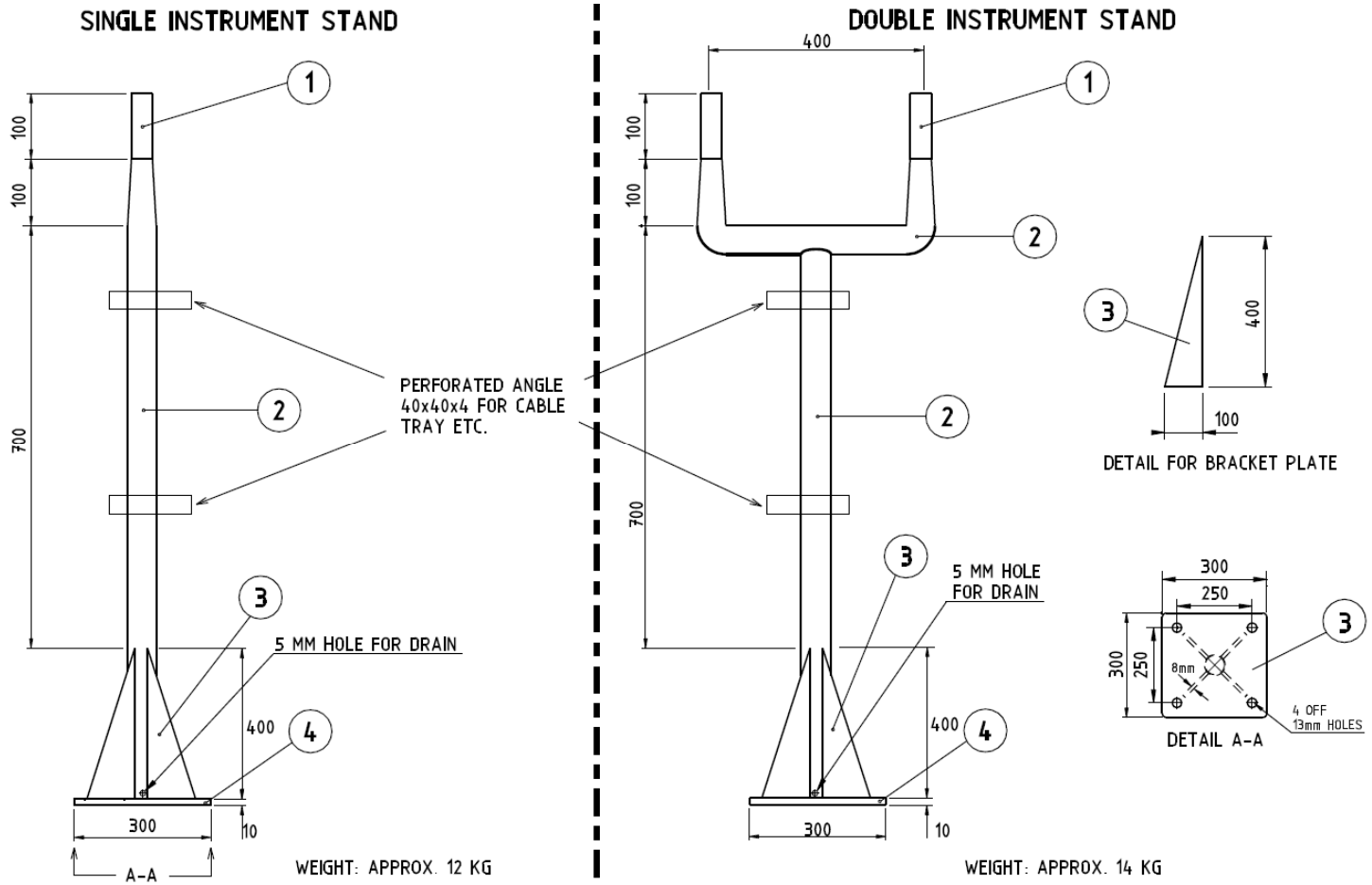
PRODUCT LIST

PRODUCT CODE	QUALITY CODE	QTY	DESCRIPTION	REMARKS

NOTES

1. MOUNTING PLATE SHALL BE SUPPORTED FROM THE STEELWORK USING THE CHANNELS, CLAMPS AND FIXINGS OF A PROPRIETRY SUPPORT SYSTEM. ('UNISTRUT' OR EQUIVALENT)

A.33 Instrument stand



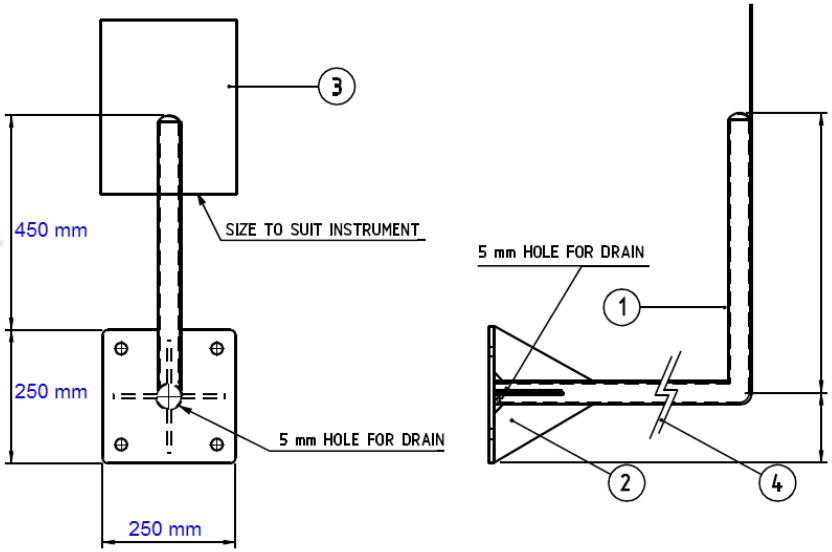
MATERIELL : STAINLESS STEEL AISI 316L

① 2" PIPE, WALL THICKNESS 2MM
② 3" PIPE, WALL THICKNESS 2MM
③ 8 MM PLATE
④ PLATE 300 x 300 x 10 MM

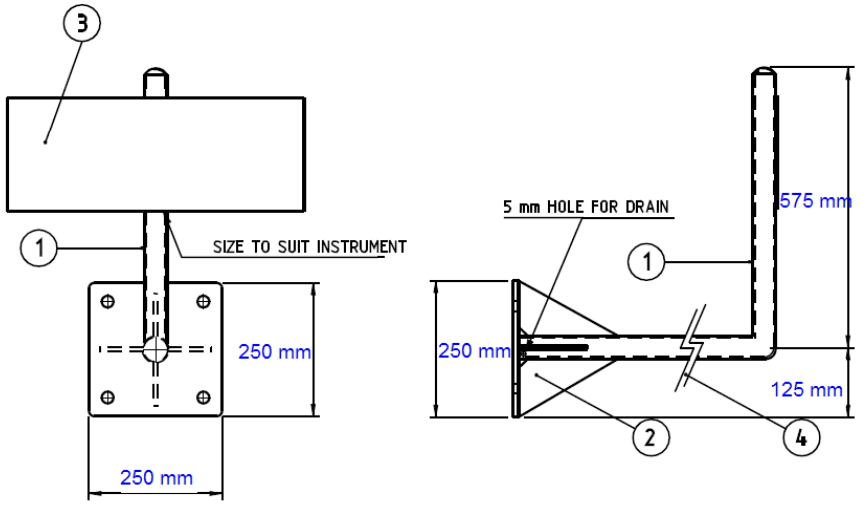
SHARP EDGES SHALL BE ROUNDED OFF
AND BURRS REMOVED

A.34 Wall mounted instrument stand with plate

WALL MOUNTED SINGLE INSTRUMENT STAND



WALL MOUNTED DOUBLE INSTRUMENT STAND



NB: ALL MATERIAL TO BE STAINLESS STEEL AISI 316L

① 2" PIPE, WALL THICKNESS 2MM

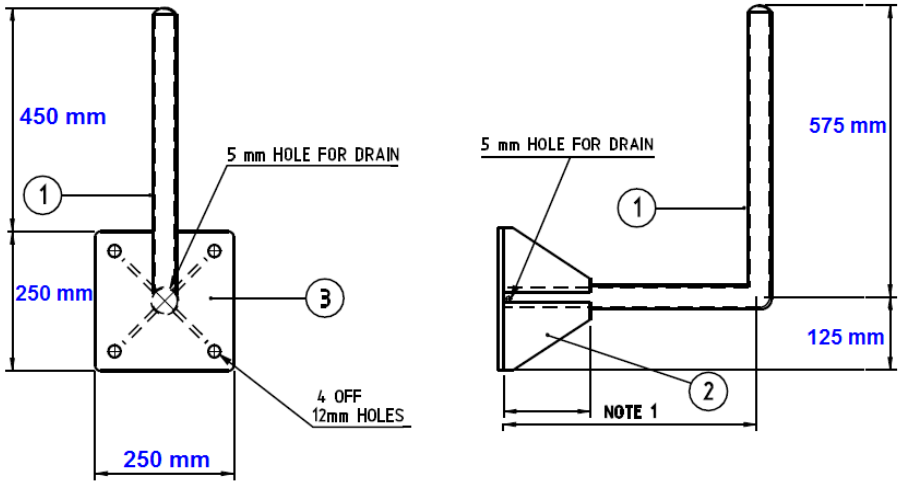
② 6 mm PLATE.

③ 6 mm PLATE FOR INSTALLATION OF F&G EQUIPMENT LIKE MANUAL CALL POINTS ETC.

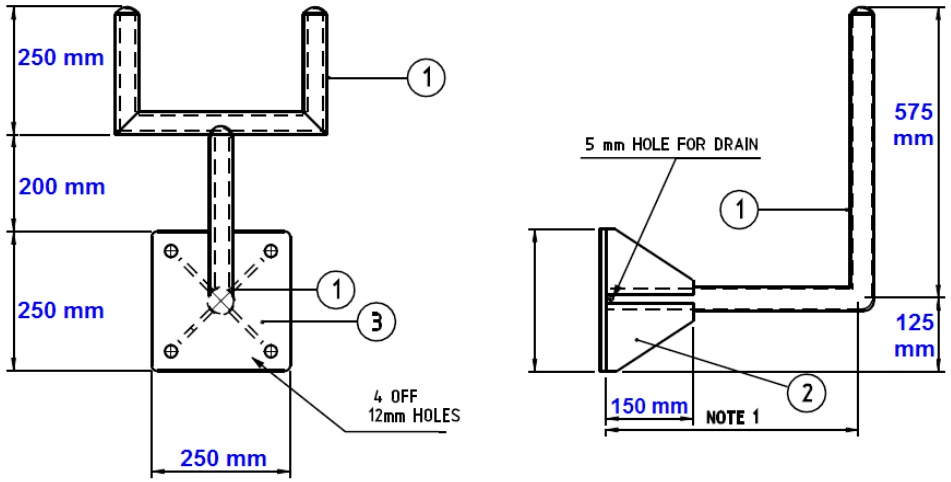
④ TO BE ADJUSTED AND PROTECTED AGAINST VIBRATIONS AT SITE.

A.35 Wall mounted instrument stand

Wall mounted single instrument stand



Wall mounted double instrument stand



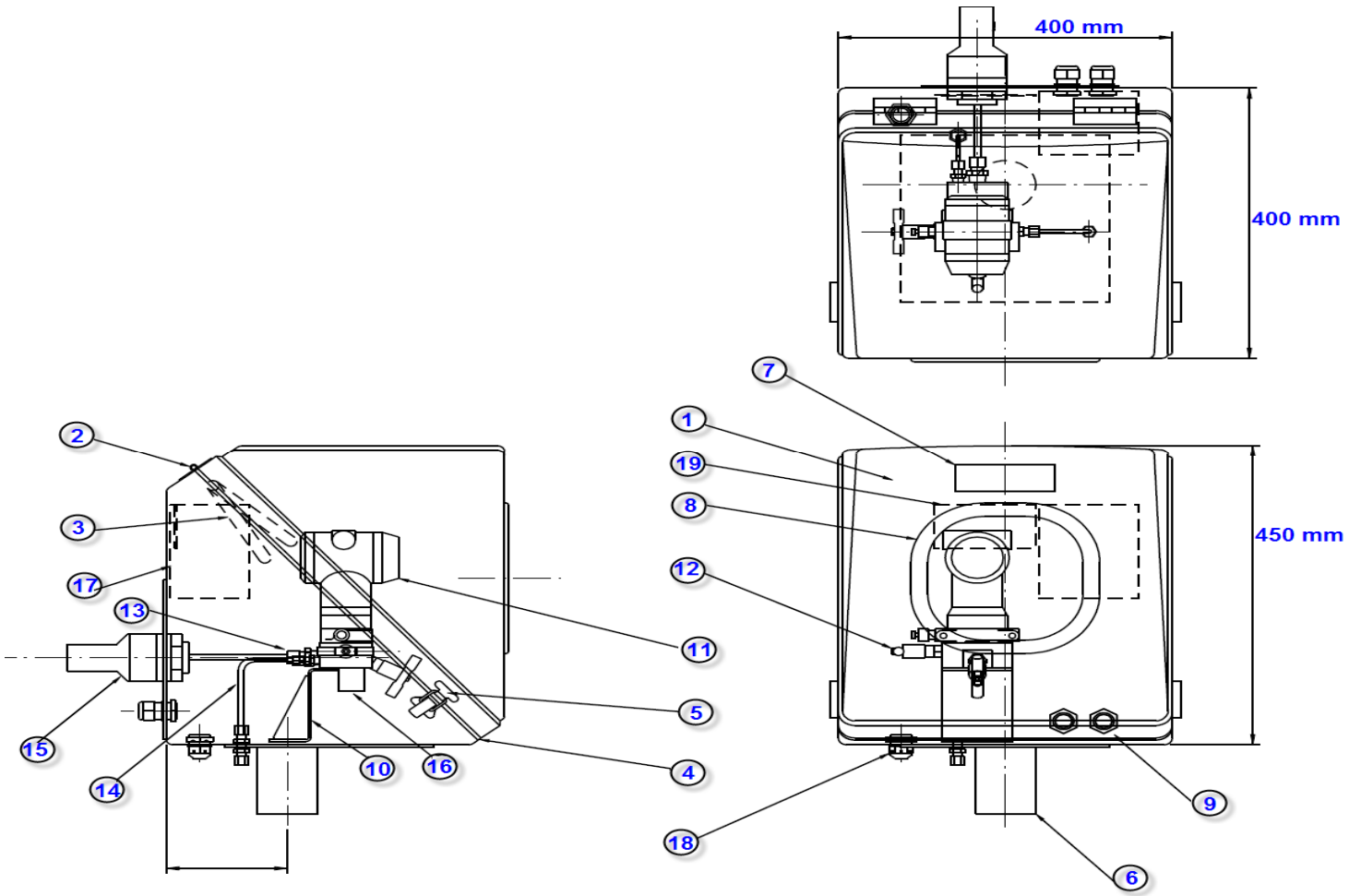
General Notes:
Alternative support solution may be used
Length to be adjusted and protected against vibration at site
All material to be stainless steel AISI 316L
Notes

(1) 2” pipe, wall thickness 2 mm

(2) 6 mm PLATE

(3) 250 x 250 x 10 mm PLATE

A.36 Instrument enclosure



ITEM	DESCRIPTION
1	GRP ENCLOSURE, MODEL 315, FIRE RETARDANT TO BS476, PART 7, CLASS 2. ANTI-STATIC TO EN50014. (COLOUR BLACK)
2	HINGE - ST. STEEL - 2 OFF
3	PROPSTAY - ST. STEEL
4	GASKET - NEOPRENE
5	TOGGLE FASTENER - ST. STEEL - 2 OFF
6	MOUNTING PLATE & CLAMP, TO SUIT 2" NB PIPE STAND - ST. STEEL
7	TAG LABEL - TRAFOLYTE W/B/W
8	VIEWING WINDOW - TOUGHENED GLASS
9	CABLE GLAND - M25 - 3 OFF
10	MANIFOLD MTG BRACKET, TO SUIT BASE INSTRUMENT MANIFOLD (ITEM 12). - ST. STEEL
11	INSTRUMENT
12	INSTRUMENT MANIFOLD - 316 ST. STEEL - 1 OFF
13	COMPRESSION FITTINGS - 316 ST. STEEL
14	6mm OD x 1.0mm WT TUBE - 316 ST. STEEL
15	TRACEPAK END SEAL - TPKES-4.
16	INST. HEATER. ST. STEEL SPRING LOADED BRACKET
17	JUNCTION BOX - 2 x M25 & 1 x M20 CABLE GLAND AND HEAT TRACE CABLE TERMINATION KIT
18	DRAIN PLUG
19	J.BOX TAG LABEL BRACKET - ST.STEEL